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FINAL DESIGN REPORT

for

Solar Powered Terrain Walker I

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EXECUTIVE SUMMARY

The challenge for our project is to develop a solar powered walking robot. It must carry a small payload, navigate obstacles, and move on terrain at an incline. Our machine will also be entered into a race against another walker under the same constraints. These objectives and constraints are the starting place in our design process for SolVoyager, our solar power terrain walker.

SolVoyager is a standard rover robot that aims to minimize risk factors in search and rescue. SolVoyager is under 18 inches so it can maneuver into small spaces and the solar power means it can go where there is no power. Previous teams attempting this project approached a design with 6 legs with 3 motors each, similar to that of an insect. It was operated on 1 Arduino circuit board. Our first conceptual design was to try recreating this insect model, attempting a 4 insect model, or make a 4 leg puppy robot. Our goals were maneuverability, balance, and design simplicity. We determined that a 4 leg puppy robot best meets these goals as it would offer the most maneuverability and design simplicity.

The final leg design is a Chebyshev 4 bar mechanism and a pantograph. A Chebyshev 4 bar mechanism takes rotational motion at the crank and translates it into rocking at the rocker. Our design generates a motion path from an extended link at the coupler link. This gives us the motion we need to walk but small and upside down. Next we use a pantograph, or scissor mechanism, to invert and enlarge our walking motion. The path generated is a semi circle where 68% is contacting the ground and 32% is the path returning along a circular arc. Simulation tools such as ANSYS help decide materials and joints as improper material and joints will result in deformation and failure.

Next, we investigate electrical components. A 20 Watt 12 volt solar panel will power the walker. As there is only 1 motor for each leg, this makes the power constraints simpler. The solar panel connects to a 5200 mAh 7.4 volt lithium polymer battery which in turn connects to each Arduino controller. The Arduino connects to servo motors that power the legs and steering motor. Servo motors were the best fit because they offer high torque and precise control. These electric components will allow us to have the power and control SolVoyager needs to complete its objectives.

The SolVoyager also needs to steer. We decided to use a 4 bar mechanism similar to that of a rack and pinion. This means the two side links rotate around the point they share with the fixed link. We can use vehicle dynamics to design it such that we can turn at certain angles. Just as for the legs, we can use ANSYS to help decide materials and map joint links. In summary, SolVoyager uses 4 legs that use a Chebyshev 4 bar mechanism and a pantograph to achieve a walking motion and steer with a 4 bar mechanism. The system will be powered by a 20 Watt 12 Volt solar panel and the motor will be led by an Arduino.

The manufacturing process for SolVoyager will involve many different machining disciplines. The 4 bar steering mechanism will be 3-D printed and connected to the plastic frame of the front legs. The legs will be a combination of 3-D printing and machining. The joint that faces the most stress will be aluminum and be machined on a vertical milling machine. The other joints will be 3-D printed. All the joints will be connected through a system of ball bearings and bolts to allow for smooth motion. All of the electronics will be ordered and assembled to power the various mechanisms.

1. PROBLEM STATEMENT

As automation takes over our everyday lives, and the need for robotic replacements to human forces in dangerous areas increases, there is an untapped industry that we aim to tackle. We introduce to you: SolVoyager, a solar powered walking robot that is capable of navigating through rough terrain autonomously. Our project goals are to design, develop, and manufacture a solar powered walking vehicle that is capable of carrying an 8 ounce payload, with potential for larger payload. The SolVoyager aims at a market for replacing the human danger factor in the search and rescue industry, as well as utilizing a photovoltaic solar panel to obtain renewable energy from the sun. With projected costs being below $700, our hope is to bring down the cost of labor and the casualties associated with dangerous environments. Some design constraints include a height no taller than 1.5 feet, autonomous directional change, and at the end of the project, outperform the second team's creation in speed and reliability.

To achieve these goals, we have devised a novel approach that combines the advantages of 4 legged locomotion, renewable solar power, and autonomous control via Arduino. Unlike wheeled or tracked robots, legged robots can traverse complex terrains and overcome obstacles that would otherwise hinder or completely stop their mobility. Solar power enables the robot to operate for long periods of time without the need for frequent recharging or battery replacement, similar to the mars rover. Additionally, onboard digital control allows the robot to adjust its pitch and direction according to the environment around it. Our robot was inspired by the popular Boston Dynamics agile mobile robot, Spot. However, our robot is much cheaper, and is capable of carrying a payload that can be used for various purposes, such as sensors, cameras, or communication devices.

In this proposal, we will describe the design specifications, physical equations, material properties, and testing methods of our robot. We will also present a detailed budget and timeline for the project, as well as a risk analysis and contingency plan. We will demonstrate how our robot meets the requirements and expectations of the search and rescue industry, as well as the potential applications and benefits of our robot in other domains. We believe that our robot is a feasible, innovative, and cost-effective solution that can revolutionize the field of solar powered walking robots.

2. CONCEPTUAL DESIGN

The objective of this project is to produce a small walking vehicle powered by the sun. Wheels, defined as a revolving object with a constant point of contact with the surface on which the vehicle moves, are prohibited. This includes treads like those of tanks or construction vehicles. It must be self-sufficient to a degree and be able to navigate obstacles autonomously. The size is a limiting factor as it must remain 18 inches or less in height. Width and length have no such constraint. It is still beneficial to keep the overall dimensions of the vehicle within reason. In a disaster relief scenario, the spaces that the vehicle must navigate through may be small and irregular. By keeping both length and width as low as possible, the walker will be able to move through a much wider array of environments. It must be able to move on its own as well as carry an 8 oz payload without any significant impact on performance.

These primary objectives introduce an issue with stability for the vehicle. In order to fulfill the requirements of the project, the vehicle must remain stable during locomotion. If the vehicle sways too much it may tip over, damage internal components, or decrease the efficiency at which the solar panel(s) charges its battery. As an additional objective determined by the group members internally, it is preferred that the vehicle's speed be as high as possible while remaining within the given constraints. These outlined constraints and objectives will produce a highly adaptable and versatile vehicle useful for a variety of tasks. As discussed with Dr. Basily, reliability is also a major factor that must be considered during design. A vehicle that breaks or falls apart easily will not be able to survive the hostile environments that disaster relief will present.

With these objectives in mind there were multiple proposals for the overall body plan of the vehicle. Because the legs are the form of locomotion, the major factor being considered was the number of legs and their orientation. Much of the electronic components remained the same between each proposal but how those components were used varied. Additionally, the design of the chassis was also determined by the form the legs took.

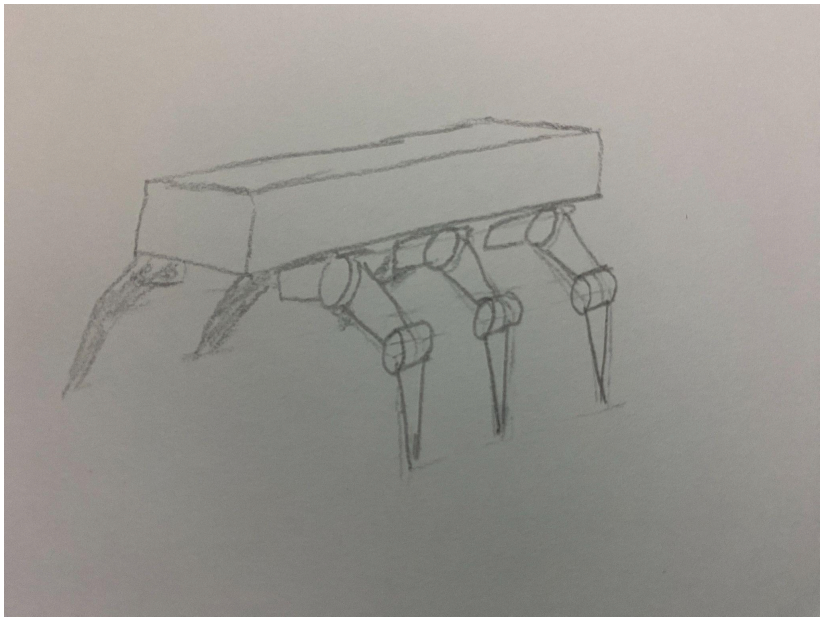


Figure 1. Design A: 6 legged Insect

The first option explored was a vehicle with 6 legs, three on each side. Each leg is articulated with three motors to provide enough dexterity for maneuverability. 2 motors attached to the base of the leg provide horizontal and vertical movement and an additional motor acts as a sort of knee joint. The legs splay outward in an insect like orientation to prevent the legs from colliding with one another during movement.

This is a common body plan in walking robots and maximizes stability and carrying capacity. The highly articulated nature of the legs allow for a high degree of maneuverability at the cost of complexity and speed. With a total of 18 motors, the vehicle would require a large battery to provide any significant amount of operating time and the accompanying solar panel would need to likewise be larger to charge it. Larger and heavier components would limit the vehicle's speed and limit the environmental conditions it can operate under.

Using 18 motors for locomotion also leads to higher mechanical and programming complexity. This would lead to lower reliability and an increased likelihood of failure during harsh conditions. Despite its drawbacks, however, the overall dexterity and stability are very important and thus the 6 legged insectoid robot remained a consideration.

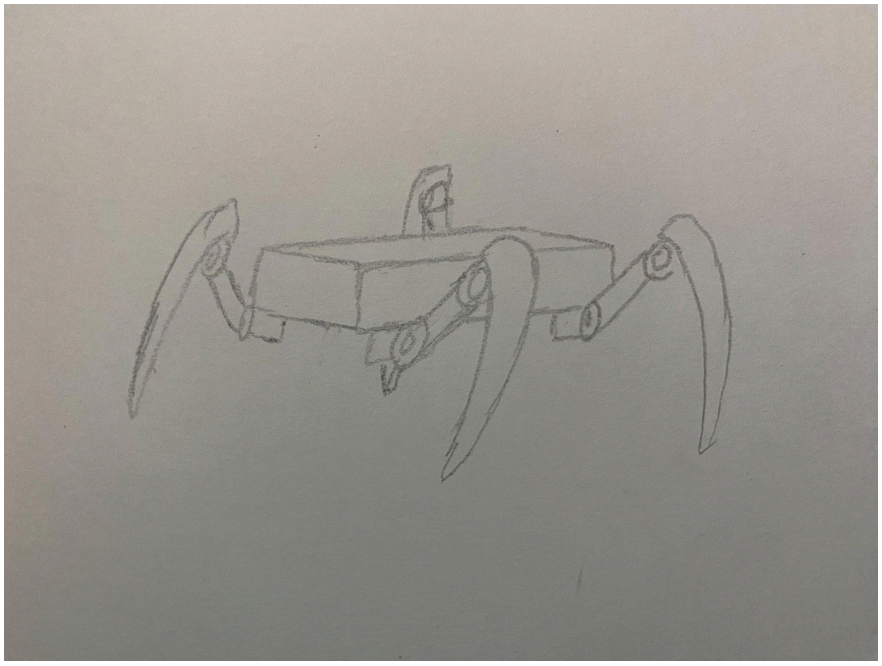


Figure 2. Design B: 4 Legged Insect

By simply removing 2 legs from the 6 legged insectoid robot, many of its drawbacks can be remedied. 18 motors becomes 12, a dramatic decrease in complexity and thus greater reliability. The downside to downsizing, however, is a decrease in stability. Because the legs are protruding outwards from the body, during the process of taking a step, the robot may become highly unstable. The best way to keep the robot stable during movement is to synchronize diagonally opposing legs with one another so that one leg can counterbalance the other. This would require very accurate servos and very precise machining of parts. Electronic gyroscopes could also be used to measure any tipping that may occur and allow the robot to counteract.

Maneuverability and dexterity may also be affected in this body plan. With 6 legs the robot can have 2 legs raised at any time with no problems with instability. With a 4 legged

design, it can only raise 1 leg before becoming unstable. This gives the 6 legged design an advantage in maneuverability as it can move more freely and perform more actions at one time without needing to worry about tipping over.

With a decrease in the number of legs, the total carrying capacity would likewise decrease. These disadvantages outweigh the benefits of such a design and thus it was deemed unworthy of consideration early in the design phase.

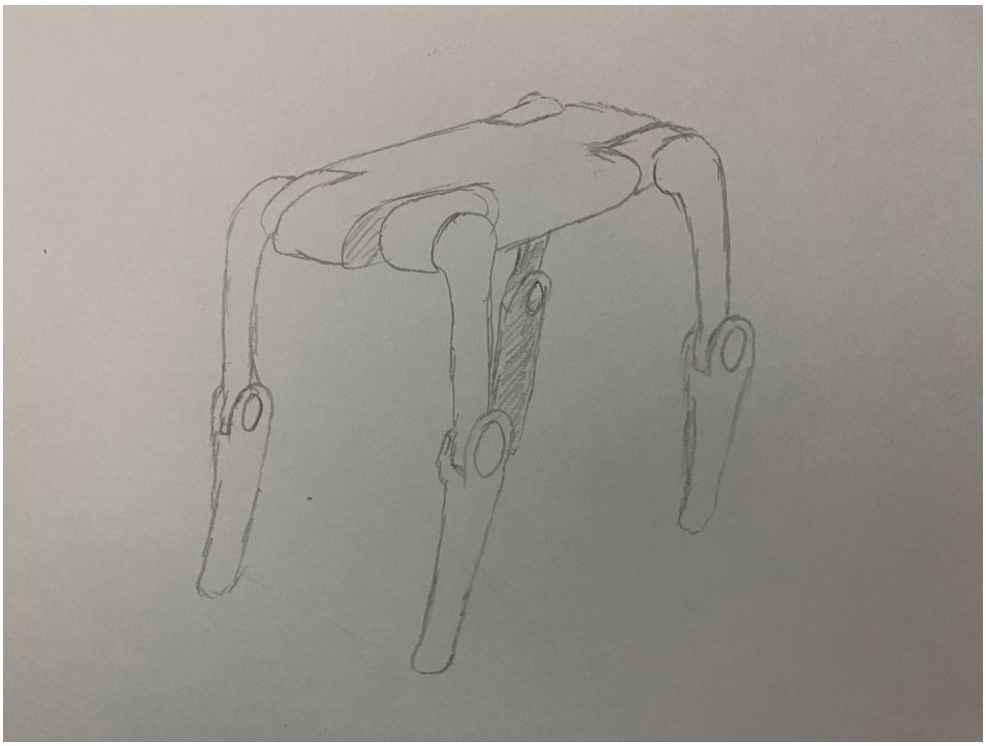


Figure 3. Design C: 4 Legged Puppy

The puppy robot remains 4 legged however their orientation is changed. Unlike the previous two options where the legs are spread outward from the body like an insect, in the puppy model the legs are vertical underneath the body. In this model the legs are better able to support the weight and are more stable than the 4 legged insect during the process of taking a step due to the legs being much closer to the center of gravity. When one leg is raised, the other three still support its center of mass.

An added benefit for the puppy leg orientation is that the number of motors can be dramatically decreased to 1 per leg. Using a Chebyshev mechanism, a single motor can operate a leg and provide enough cyclic horizontal and vertical motion to move the robot forward and backward. Changing direction can be accomplished through a separate system discussed later. Limiting the number of motors to one per leg dramatically decreases the power draw and will allow for a greater operational period. Alternatively, using a smaller battery may be useful as it would decrease the overall weight. While it seems this model is the perfect cross between stability and complexity, there are some drawbacks.

Because the leg assembly is entirely vertical and contains unanchored joints, the legs may be susceptible to lateral forces. As such, the material composition of each leg segment and joint must be strong enough to not sway side to side. Stronger materials are typically heavier and/or

more expensive so budget concerns may arise. Additionally, because the single motor only rotates about one axis, the legs are incapable of motion perpendicular to the direction of travel. This will limit maneuverability greatly and may impact how well the vehicle performs on tough terrain. Despite these drawbacks, however, the design's offerings in terms of reliability and stability stand out and therefore this will be the prevailing conceptual design.

Chassis

The goal of the chassis is to have a stable frame for our legs and components and have the ability to steer what direction we are going in. The legs will be mounted on the sides of the chassis. The solar panel will be mounted on the top of the chassis. All other electronic components will be placed inside the chassis. The goals of our chassis are directly related to the legs that we must pick for our project. For insect legs the chassis can be a solid frame with no degrees of freedom. However, a design with legs like a dog would require extra attention for steering. In order to steer with a chassis with legs like a dog the chassis would need at least 3 degrees of freedom. The motion would require 1 rotational and 2 translational degrees of freedom. A chassis that describes this degree of freedom requirements is a car. So a chassis that can accomplish these goals is either a fixed chassis with insect legs or a chassis with degrees of freedom similar to a car.

Since we are using a design with 4 legs in the style of a dog, we must refer to the design of a car for our starting concepts. There are many different ways that vehicle turning can operate. However, all of them operate on the same principle. That principle is that linear motion combined with a change in steering angle result in motion along a fixed curve. The path that a vehicle travels when steering is referred to as radius of curvature. This is important in vehicle dynamics because the forces experienced by the chassis are impacted by the radius of curvature. The following equation is taken from the book "Applied Dynamics" by Prof. Haim Baruh, a textbook on vehicle dynamics.

$$a_y = v_G^{\,2}/R_S \qquad [1]$$

Here we can see that as the radius of curvature increases, the acceleration in the lateral or y-direction increases. When discussing stability of a chassis we must take into consideration the turning radius. This will be a concept discussed in more detail later in our report. For now just remember that turning causes instability and our job is to design a chassis that can handle it. Incorporating steering into our chassis is important and understanding the fundamentals of a car's chassis will help us do that.

To reiterate, turning is a combination of linear motion combined with a change in steer angle that results in motion along a fixed curve. Vehicles have a few options when it comes to accomplishing this goal of turning. The first is how cars turn. Cars operate by rotation of the wheels along the z-axis or vertical axis. Cars have 4 wheels, two front and two rear. Cars can operate either by having just the two front wheels and turning or by having all 4 wheels turn. The principle of how much the wheels turn by is based on the radius of curvature concept explained above. All the wheels should be on a path where the center of the radius of curvature is the same. In a car in which the back wheels do not turn they are already aligned since they are on the same

axle. The front wheels are turned at different angles so that they match up with the same radius of curvature. This motion requires the inner and outer wheels of the car to be moving at different speeds to allow for the same turning radius. The inner wheels would be traveling slower than the outer wheels. A vehicle model based on a car's steering is one possible option for designing our chassis.

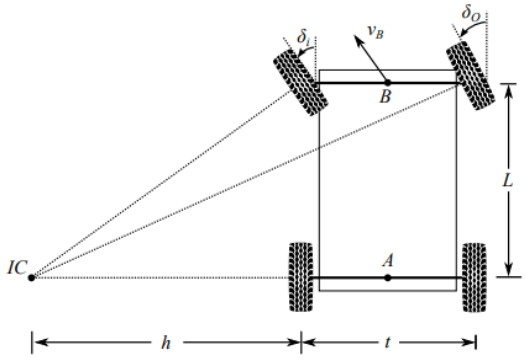


Figure 4. Diagram from Applied Dynamics showing Steer Angle and Instant Centers (Ref. 8)

There are many benefits to car steering design but they come with their own set of challenges. Car steering allows us to have one stable main chassis and 2 legs attached such that they have 3 degrees of freedom. This would make balance of the robot and stability easier. The car steering would also allow for a lot of control in the angle we turn at. There are also many resources available to better understand car steering in our future analysis. Some of the challenges of car steering are getting the legs to move at different speeds. If we lose constant contact our robot will fall over. We would need to adapt the robot's walking pattern to allow for turning. The car steering should be possible but our detailed analysis will determine if we can overcome its challenges.

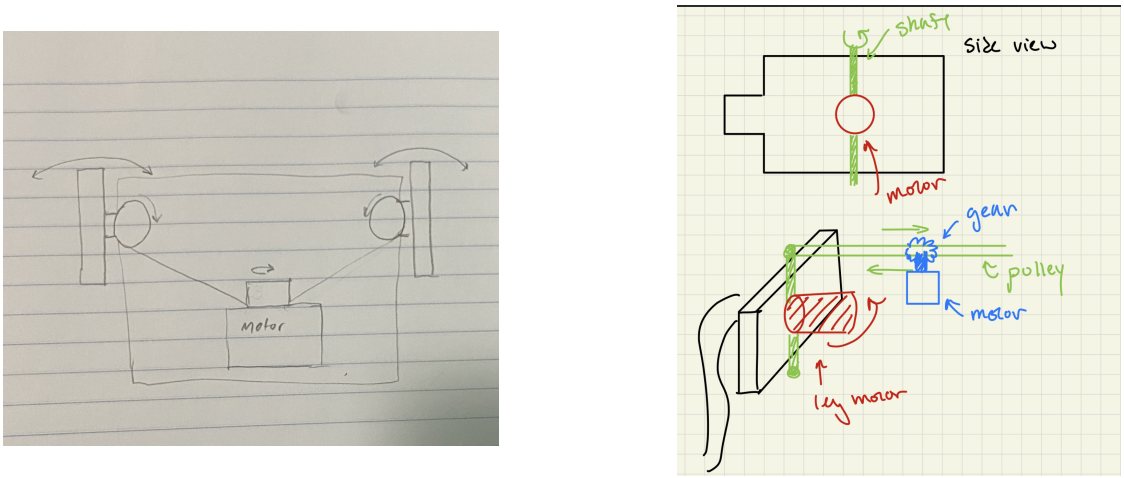


Figure 5. Car Steering Chassis Design

Another option that vehicles use to turn is steering that rotates the front half of the chassis from a central pivot point. This is not seen in cars but seen in construction equipment. The front wheels and the back wheel are aligned such that the inner and outer are always at the same angle. Our design will not be going at the high speeds of a car so in many ways this design reflects the

goals of our robot. There are many ways we could accomplish the rotation of the center of the chassis. A hinge could connect the front and back chassis. A motor and gear can be attached to the chassis and hinge. This would give our robot the 3 degrees of freedom needed to turn. In this design we could synchronize the legs on the same motor. This way the only synchronization would be between the front and rear instead of each leg separately.

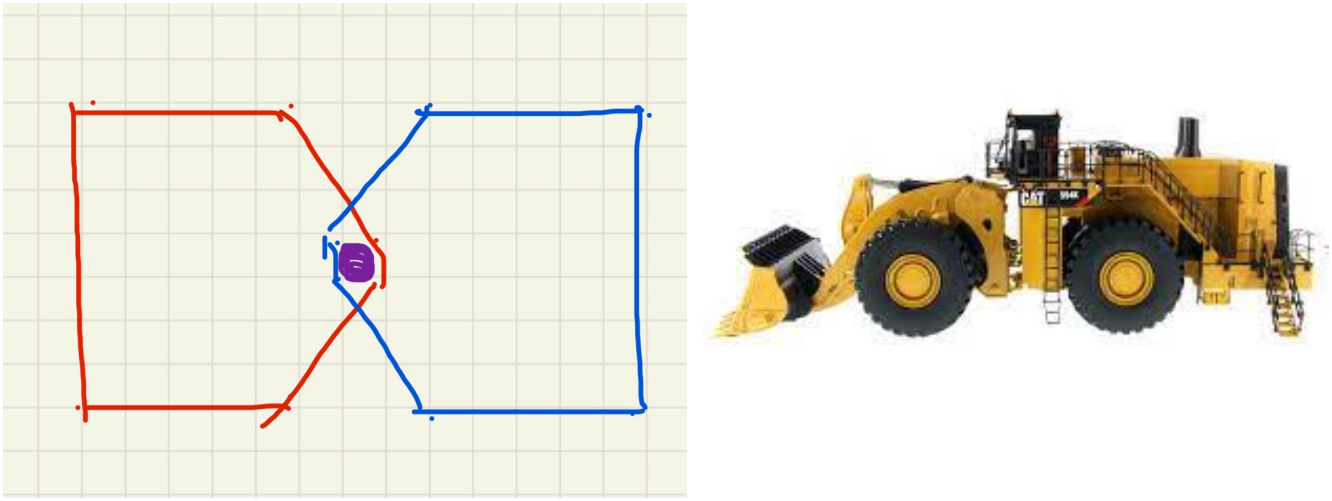


Figure 6. Center Pivot Point Chassis Design

The center pivot steering offers many new advantages but comes with some drawbacks as well. The first advantage is having more control over the angle we steer at. This would allow us to have more precise navigation. Another advantage is that we could synchronize both front legs on the same motor. This would reduce the amount of motors required. Also, the synchronizing of these legs would make walking smoother. Even the setup of the center hinge with a single motor would be easy to design. There are some disadvantages when separating the chassis. One disadvantage is balancing the weight distribution. In the previous design the weight is all in one chassis so the legs would all be experiencing the same loading. With a chassis in two parts we would need to make sure that both parts of the chassis weigh the same amount. Also since we are turning the entire front of the chassis there is a large lateral force. This would mean our legs would need to be designed to stay stable and balanced even when we are turning. Center pivot steering offers different problems from our first concept but still satisfies our need for steering.

We brought both these ideas to our feasibility meeting with Dr. Basily. He found both the ideas regarding turning as possible. He described to us in great detail how we could accomplish these ideas better. We will talk about each chassis design separately. First we will discuss the car style steering. Originally we were going to control the angle we were turning at with two steering motors one for the left and right front legs. We had also considered using a rack and pinion and having one motor control the steering. However, Dr. Basily suggested an even simpler idea. We would have a 4 bar mechanism with two fixed points. One of the bars would be connected to a servo motor with an arm. The arm would have a slider on it. This slider arm connected to the 4 bar mechanism would act like a rack and pinion but we would not need to do much of the math that goes into designing. These ideas make much of the challenges we were facing simpler.

The central steering concept was also discussed during this meeting. He found both the ideas regarding turning as possible. He gave us suggestions on how we should attach the hinges so that the weight on the two chassis parts have an even weight distribution. Dr. Basily described the setup as having a top and bottom hinge. One chassis's hinges would be squeezed inside the other hinges. This would create a torsional force but the weight would be evenly distributed onto both sides. And the hinges would be metal and would have no problem withstanding the torsional forces. This design would remove the balancing challenge outlined before. However, we would still need to adjust to the strong swing force which may cause our robot to lose balance. Overall this design was greatly improved by the meeting with Dr. Basily.

Once we had our meeting with Dr. Basily, we had to decide on which model would be best for us going forward. Both designs were feasible but presented different challenges. The car steering chassis offers a simple one part with separate hinges for the legs to allow turning. However, challenges with this design involve synchronizing the legs with turning a motion. The center chassis offers a simpler and more controlled way of turning. It also gives us the option of synchronizing the legs by front and rear. However, separating the chassis and moving it creates issues with balancing cause from changing centers of gravity and swinger forces. These were the problems we faced when comparing both designs.

We decided to pursue the car steering chassis. From our understanding of previous projects and discussions within the group we felt that balancing is a key issue. The center steering adds issues to steering that would involve gyroscope and balancing systems. This seemed too complicated in addition to other parts of our project. The car steering chassis will allow us to have a stable body that all our components can sit in. Then we can just pivot the leg modules and keep the body still. This makes balancing much easier. There are some challenges that go along with this design. The first will be synchronizing the legs to match our motion path. This will be done with a code solution discussed later. The second challenge will be designing the turning mechanism to move both the legs to desired turning angles. The feasibility meeting gave us possible solutions that we will explore in greater detail later. In conclusion, while the car steering offers a set of unique challenges we feel it is the best solution for turning when balancing issues are concerned.

3. DETAILED DESIGN AND ANALYSIS

Leg Analysis and Equations

The final leg design consists of a Chebyshev 4 bar mechanism and a pantograph. The 4 bar mechanism is a Chebyshev mechanism, meaning that one link, that being the crank, can make a full 360 degrees of continuous rotation. This allows us to connect a motor to the crank link and actuate the one degree of freedom of this mechanism. This mechanism is different from typical 4 bar mechanisms because the coupler extends beyond its joint with the rocker, to its end joint, all being one rigid link. The 4 bar Chebyshev mechanism is shown in 2d below, showing the motion of each joint.

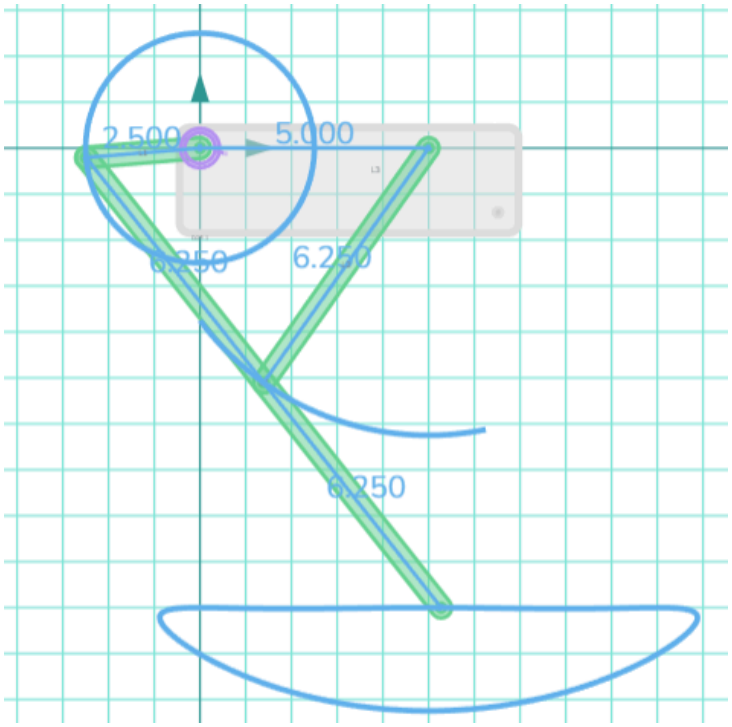


Figure 7. Motiongen 4 bar Chebyshev mechanism

Further differences with a typical 4 bar mechanism is the output. The input, that being the crank, is the same as a typical mechanism, shown in the top left of the figure it makes a full 360 degrees of rotation. Instead of the output being the rocker (like a typical 4 bar mechanism), it is the end joint of the coupler, and the rocker acts as a guide and support link. This mechanism configuration allows us to achieve the motion path of the end joint of the coupler as shown above. This motion path has a very flat portion and a semicircle path that returns the mechanism to the flat part. This mechanism cannot be used alone to actuate the robot, as the output path is upside down and small. To change this path to the desired path, we used a pantograph.

Pantographs, also known as scissor mechanisms, use parallelograms to invert, enlarge, or shrink an input path to an output path. This mechanism is an old and widely used mechanism, seen practically as far back as the early 17th century. This mechanism has wide applications, as seen in its use in sculpting, minting, milling, and many other industries. We will use a

pantograph to invert and enlarge the output path of the 4 bar mechanism such that the flat part of the pantograph's output will be the leg in contact with the ground and the semicircle path will be the leg lifting off the ground to take another step. This connection of the 4 bar mechanism and the pantograph is very specific, because if the 4 bar mechanism was connected to another joint, then the path would not be inverted and enlarged, depending on the joint it could have been shrunk.

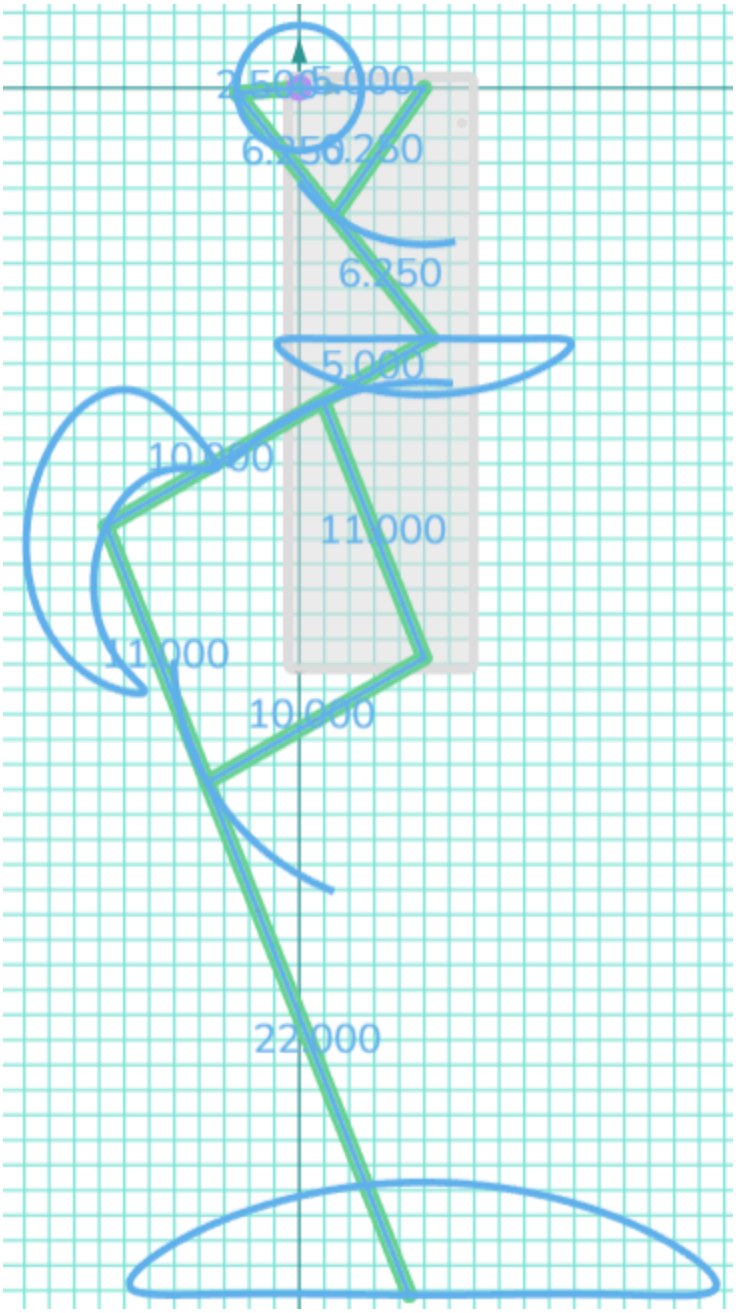


Figure 8. Motiongen final leg model

As can be seen in this figure of a motiongen.io simulation of our final leg mechanism, the pantograph successfully enlarges and inverts the output of the 4 bar mechanism. The pantograph

by nature only has one fixed joint which will hold most of the weight of the robot. With the final mechanism configuration, a stable motion path is established. This however is not the only desirable property of this mechanism. This mechanism has a quick return feature, that is that the time it takes to return the mechanism to its desired output is fast, relative to the time it spends in its desired output range. For our case, the desired output is the flat portion of the motion path, which is when the leg is in contact with the ground. The quick return feature minimizes the time the leg is not in contact with the ground. We found that ideally, with a constant rotation of the motor at the crank of the 4 bar mechanism, about 75% of the cranks rotation of the leg is in contact with the ground and about 25% where the leg is in the semicircle motion, returning to the ground. In reality, we found that this feature works even better. This is because when the leg initially is lifted off the ground, most of the load on the motor will be gone, baring the resistance of the mechanism. This will in turn make the motor spin slightly faster when it is not in contact with the ground as the constant output of the motor will have less resistance, hence more speed. This feature of our leg mechanism minimizes the time each leg spends not touching the ground, vastly improving prospects of stability and speed.

For the leg mechanism design to be feasible there are certain criteria that need to be met for it to work smoothly. As shown in the mechanism in Figure 1 it is important to make sure the linkage lengths that are chosen allow the mechanism to achieve a specific motion path at the foot (or point A) and for the transmission angles (γ_1 and γ_2) are within a reasonable range. To ensure that this is the case position analysis has been done for the mechanism.

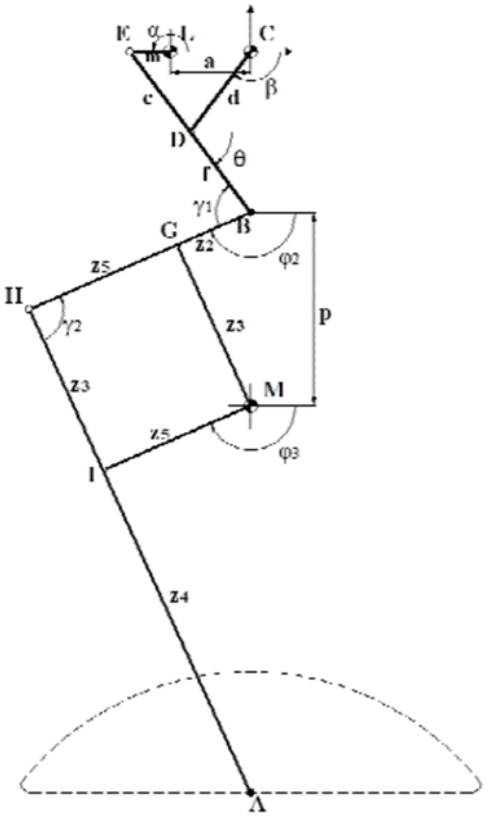


Figure 9: Leg Mechanism Schematic

For linkage names, they are read by the joints in the rigid link, from top down and right to left. Ex: crank is link LE, coupler is link EDB

Linkage lengths: (inches)

$a = 1.25$

$b1 = 1.875$

$b2 = 3.75$

$c = 1.5625$

$d = 1.5625$

$f = 1.5625$

$h = 5.75$

$m = 0.625$

$z_2 = 1.25$

$z_3 = 2.75$

$z_4 = 5.5$

$z_5 = 2.5$

$z_6 = 2.75$

$z_7 = 2.5$

Equations:

When looking at equations it is important to point out that the origin of the mechanism is the point C as seen in Figure 1. Standard notation for joint locations in this paper will follow the example of X_M, where the X axis location of the joint M of the mechanism and Y_M would represent the Y axis location of the joint M.

$\alpha = 0\ to\ 360\ degrees$ **[2]**

α is the input angle which measures the angle from the X-axis to the crank (link EL as seen in Figure 1) each calculation made after is based off of the specific angle that alpha is at.

$X_M = 0$ **[3]**

$Y_M = -5.875in$ **[4]**

The position of point M is chosen since it is a fixed point.

$X_E = a + (m * cos(\alpha))$ **[5]**

$Y_E = m * sin(\alpha)$ **[6]**

$$X_B = X_E - ((c + f) * cos(\theta)) \quad \textbf{[7]}$$

$$Y_B = Y_E - ((c + f) * sin(\theta)) \quad \textbf{[8]}$$

For Equations 6 and 7, theta is the measurement of the angle taken from the x-axis to the link EB.

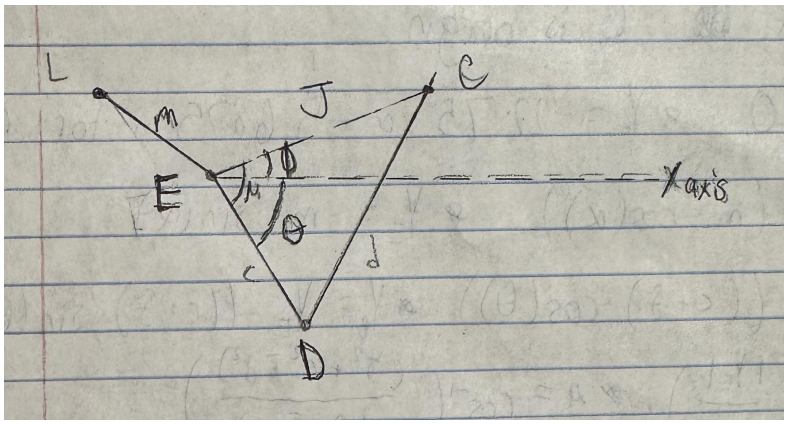


Figure 10. Leg position calculations

$$\theta = \mu + \phi \quad \{1 \leq \alpha \leq 180\} \quad \textbf{[9]}$$

$$\theta = \mu - \phi \quad \{181 \leq \alpha \leq 360\} \quad \textbf{[10]}$$

$$\mu = cos^{-1}\left(\frac{J^2+c^2-d^2}{2*J*c}\right) \quad \textbf{[11]}$$

$$\phi = tan^{-1}\left(\frac{|Y_E|}{|X_E|}\right) \quad \textbf{[12]}$$

$\phi\ and\ \mu$ are just angles found in order to get θ which is important to have in order to get the position of joint B.

$$\gamma_1 = \theta + J_{(angle)} \quad \textbf{[13]}$$

γ_1 as shown from Figure 1 is one of the transmission angles of this mechanism which is important to take note of because typical ranges for this value that are deemed acceptable is in the range of 45-130 degrees where the ideal transmission angle for a mechanism is 90 degrees. Transmission angles are important to a mechanism since good transmission angles allow force to be transmitted more effectively.

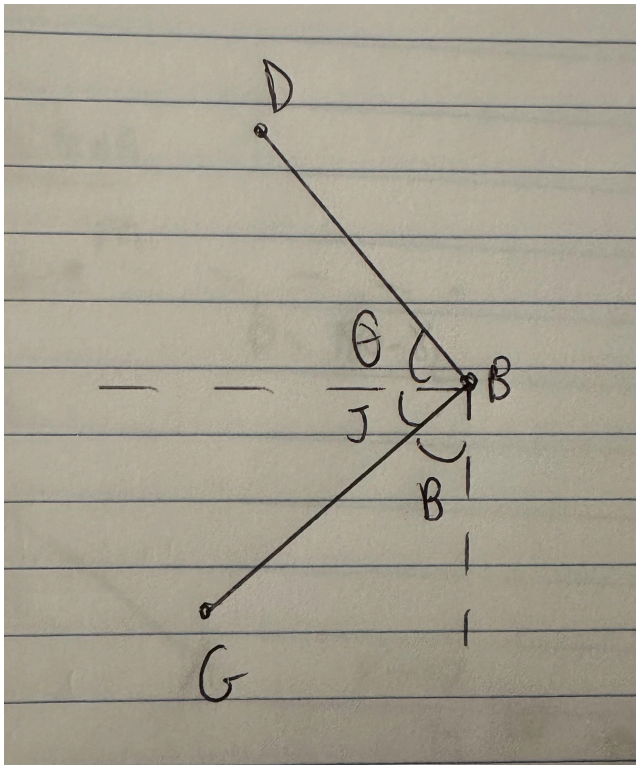


Figure 11. Leg position calculations

$$J_{(angle)} = 90 - B_{(angle)} \quad \textbf{[14]}$$

$$B_{(angle)} = cos^{-1}\left(\frac{z_2^{\,2}+D^2-z_3^{\,2}}{2*z_2*D}\right) \quad \textbf{[15]}$$

$$D = \sqrt{(X_B - X_m)^2 + (Y_B - Y_m)^2} \quad \textbf{[16]}$$

$$\gamma_2 = cos^{-1}\left(\frac{z_2^{\,2}+z_3^{\,2}-D^2}{2*z_2*z_3}\right) \quad \textbf{[17]}$$

Both graphs of the transmission angles against the crank angle are shown in Appendix C.

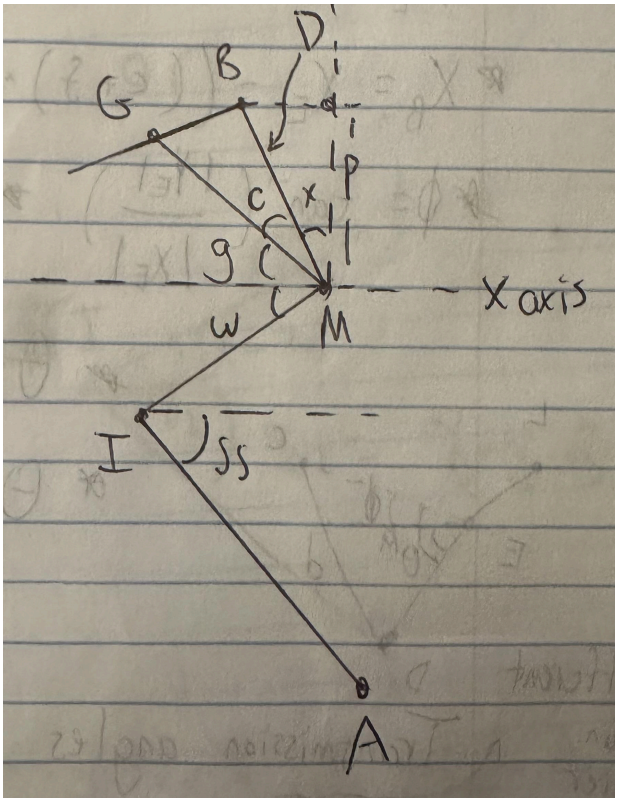


Figure 12. Leg position calculations

$$P = Y_B - Y_M \quad \textbf{[18]}$$

$$c_{angle} = cos^{-1}\left(\frac{D^2 + z_3^2 - z_2^2}{2*D*z_3}\right) \quad \textbf{[19]}$$

$$x_{angle} = cos^{-1}\left(\frac{P}{D}\right) \quad \textbf{[20]}$$

$$g_{angle} = 90 - c_{angle} - x_{angle} \quad \textbf{[21]}$$

$$w_{angle} = \gamma_2 - g_{angle} \quad \textbf{[22]}$$

$$X_I = X_M - z_5 cos(w_{angle}) \quad \textbf{[23]}$$

$$Y_I = Y_M - z_5 sin(w_{angle}) \quad \textbf{[24]}$$

$$ss_{angle} = \gamma_2 - w_{angle} \quad \textbf{[25]}$$

$$X_A = X_I - z_4 cos(ss_{angle}) \{1 \leq \alpha \leq 180\} \quad \textbf{[26]}$$

$$X_A = X_I - z_4 cos(ss_{angle}) \{181 \leq \alpha \leq 360\} \quad \textbf{[27]}$$

$$Y_A = Y_I - z_5 sin(ss_{angle}) \quad \textbf{[28]}$$

Point A represents the foot of our robot so analytically being able to show that the motion path generated by this mechanism is what is needed is very important. The graphical representation of the motion path created by points A, E, I and B are shown in Appendix C Figure 4.

In the graphs of the transmission angles verses the crank angle it was shown for γ_1 to have a range of $83^\circ \; to \; 153^\circ$ and for γ_2 to have a range of $80^\circ \; to \; 128^\circ$. The first transmission angle does have a higher transmission angle then what is ideal however in motion simulations it was shown that this won't have a significant effect in the performance of this mechanism. The second transmission angle is within the traditional range. Ideally, transmission angles are best kept at 90 degrees, however it is practically impossible to maintain that transmission angle. A general rule of thumb according to Kinematics and Dynamics of Machinery, 3rd Edition is the absolute value of any transmission angle minus 90 degrees should be less than 45. In other words, the transmission angles should be kept within 45 degrees of 90 degrees. This is true for γ_2, however there are a small range that γ_1 does not fit this criteria. Despite γ_1 not always conforming to this criteria, it is overall still a good transmission angle because of the short duration it does not satisfy the criteria.

Now that we had determined the final 2D configuration of our one degree of freedom leg mechanism, we set out to design the 3D aspects of the leg. The leg mechanism has a total of 3 fixed joints, 2 for the 4 bar mechanism and one for the pantograph. For the 4 bar mechanism, the fixed joints are at the same altitude, 1.25 inches apart. The left fixed point is the motor's output, which is attached to the crank. The pantograph's fixed point is 5.6875 inches below the right fixed point of the 4 bar mechanism. Naming these fixed points for reference, the 4 bar mechanism's left fixed point is referred to as the motor output, the right fixed point of the 4 bar mechanism as the upper fixed point and the pantographs fixed point as the lower fixed point. Based on the location of the fixed points, we decided that the base that holds together the leg unit will be a vertical beam for the upper and lower fixed point, and a platform will be attached to the beam such that the motor can be attached to serve as the actuator and 3rd fixed point. This base will be attached to the chassis via pivot joints for the front legs and directly attached for the back legs, which will be expanded on later. The upper and lower fixed joints will have M3 holes tapped into the base, where shoulder screws with 4mm diameter shoulders of various lengths are inserted after being threaded.

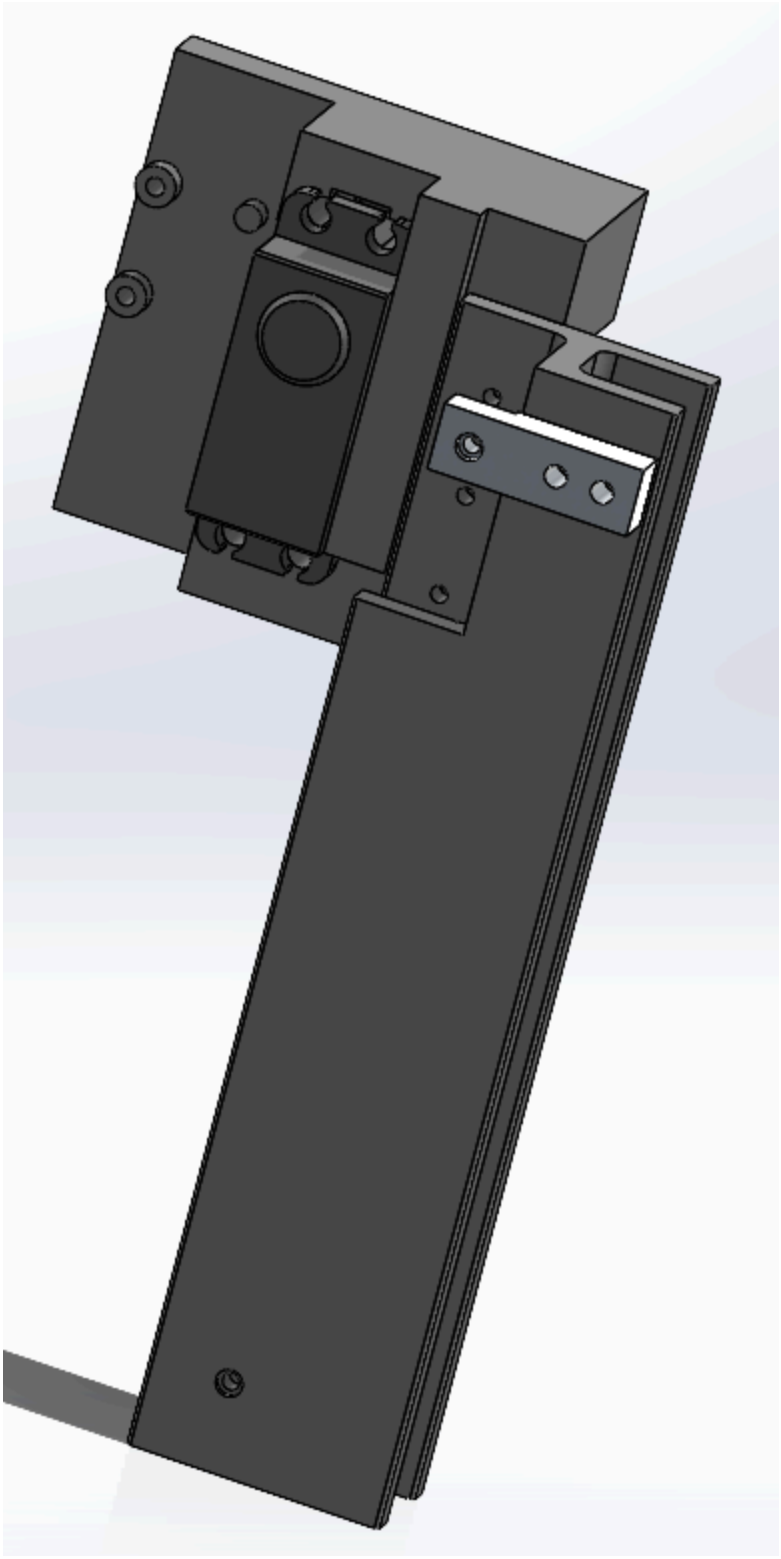


Figure 13. Leg base with motor and M3 holes for fixed joints

To achieve the desired motion path set out in 2D into 3D requires the links to be side by side with each other. A problem with simply connecting one simple link to a geometrically

similar link side by side is the shear stress and bending moments when force is applied at the joints. These factors can lead to the shaft and or ball bearings shifting and generally forces acting in directions they should not. To avoid this, the links composing the mechanism are classified into 2 types, inside and outside links. At moving joints the outside link is on either side of the inside link, securing the shoulder screw with a 4 mm diameter hole on the outside and threaded M3 hole on the other side. Two set screws are used to secure the shoulder screws in the outside link on the side of the link without the threads. The outside link is connected with the other side of itself outside the range of the inside links rotation, making it rigid. The inside link has a deep groove ball bearing for the joint that finishes the joint connection. Our previous design of the joint used ball bearings for both the inside and outside links with a bolt and nut to secure it, totaling 3 bearings per joint. This final configuration only uses one ball bearing and requires no nuts and less washers, reducing both weight and cost. The washers are used between both inside walls of the outside link and the ball bearing of the inside link such that the inside link is not rubbing against the outside link.

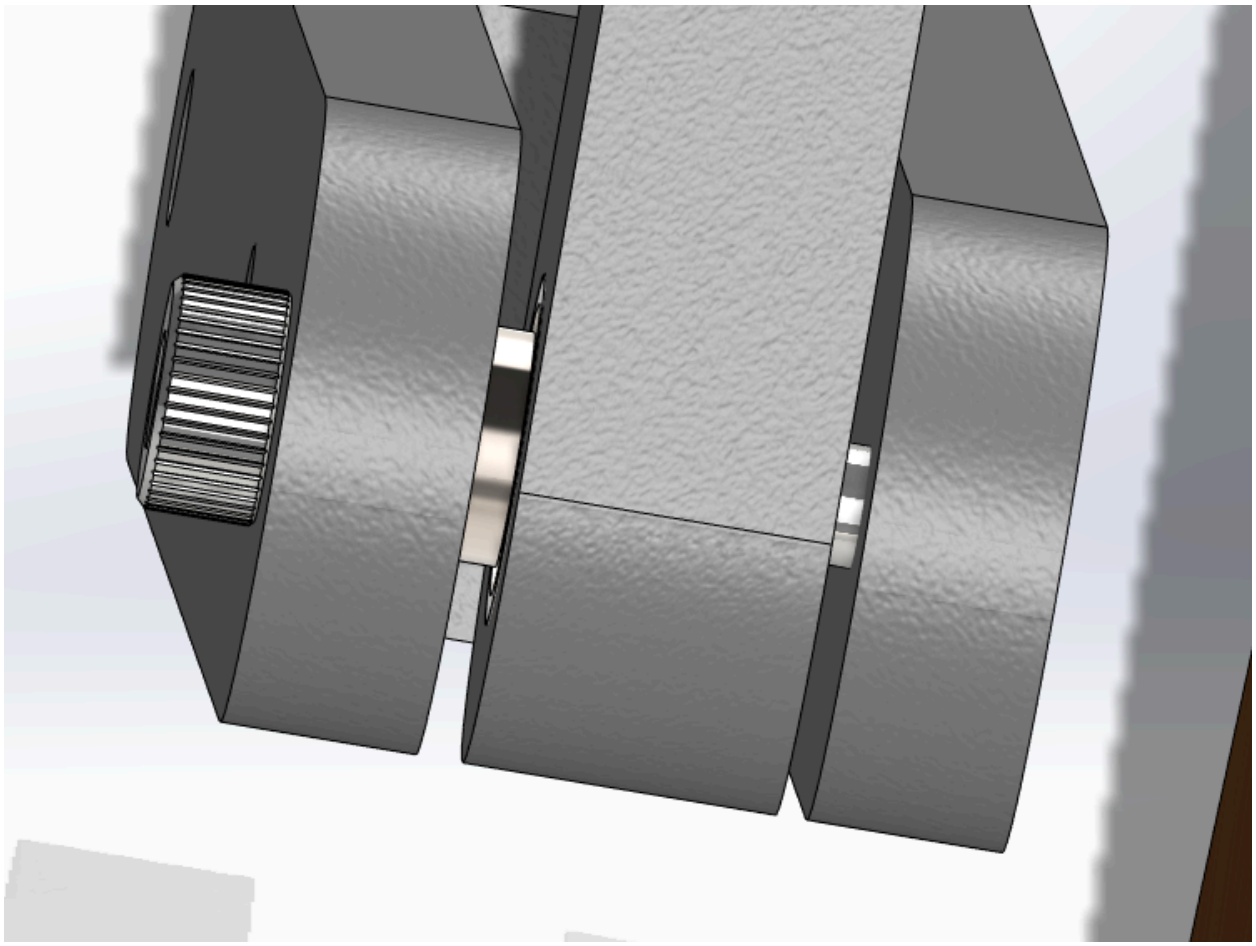


Figure 14. Joint B close up; showing inside outside joint configuration

There is one exception to the default inside outside joint configuration, that being the cranks connection to the coupler of the 4 bar mechanism, link EDB from figure 1. At this joint link EDB is an outside link and because the crank needs to make 360 degrees of rotation, the crank cannot go in between the EDB link. This requires the side by side connection with a bolt

securing the shoulder screw in, which holds in the 3 ball bearings. The motor will be screwed into the stand and a 25T servo horn will be attached to its output. The two screws will be screwed into the crank, both holding it up and actuating it.

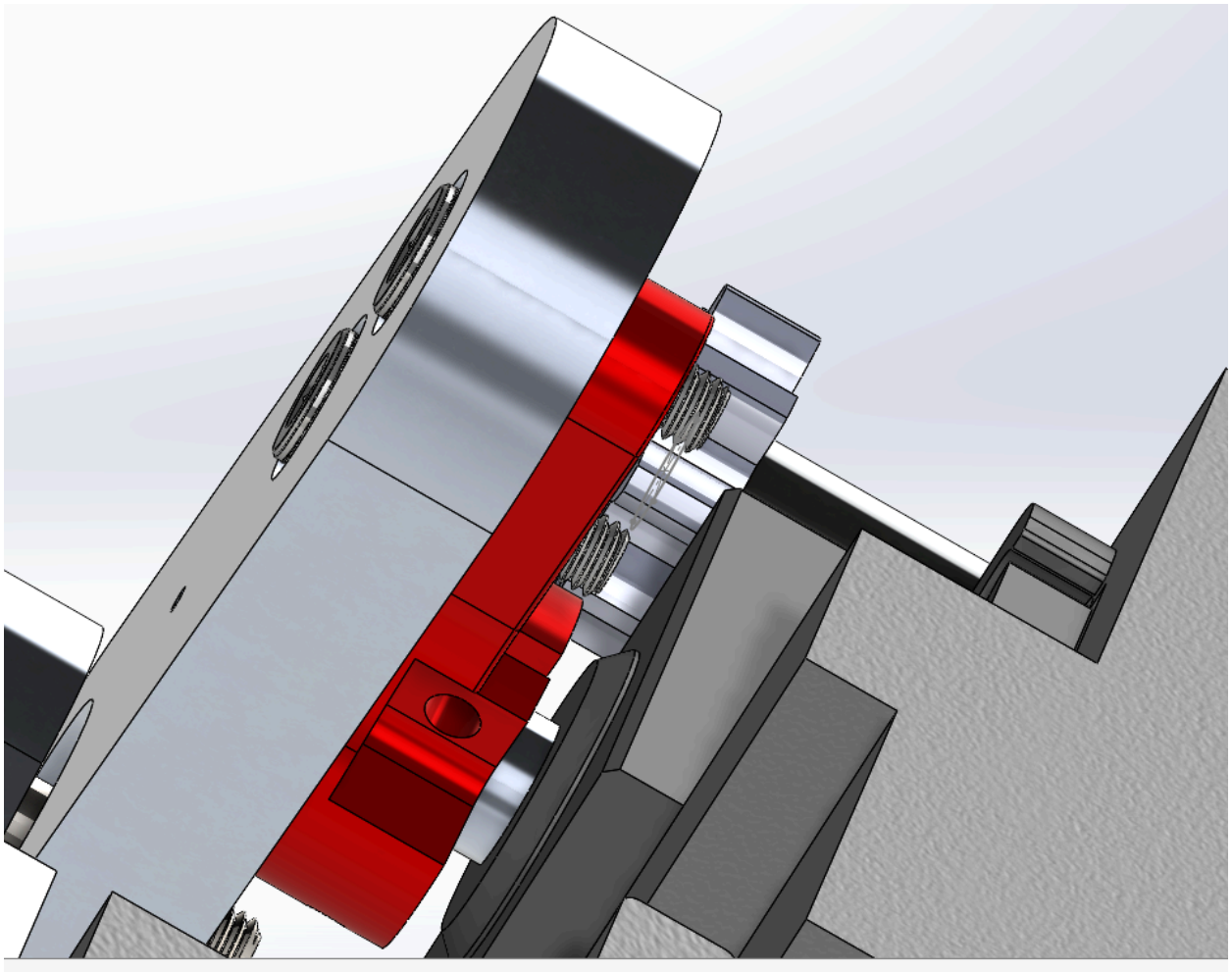


Figure 15. Servo Horn connection with crank without bolts

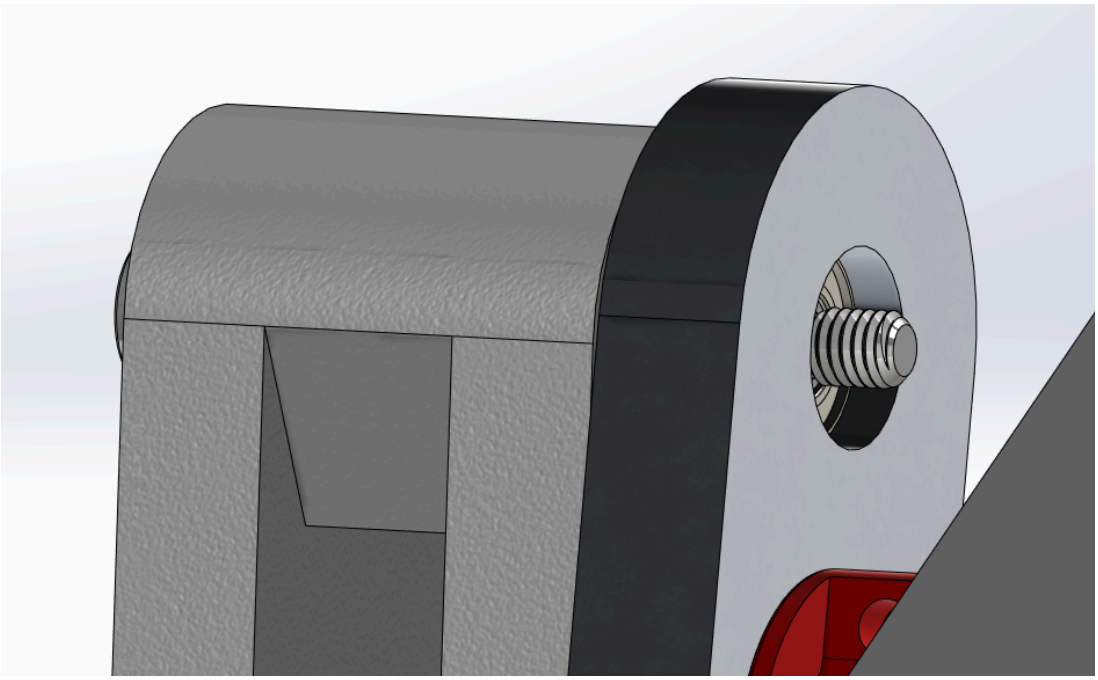


Figure 16. Joint E Without bolt

The fixed joints are similar to the moving joints, except in size and they have to all use ball bearings. The base will have M3 screw holes tapped and threaded in the exact locations of the fixed joints. The 4 mm shoulder screw will be screwed into these holes after being inserted into their respective ball bearings. We found this is not only better than carving the shaft from the base, but also better in keeping the bearings in the same place. This is because the shoulder screw once screwed in will apply some compression with the washers, preventing the bearings from shifting.

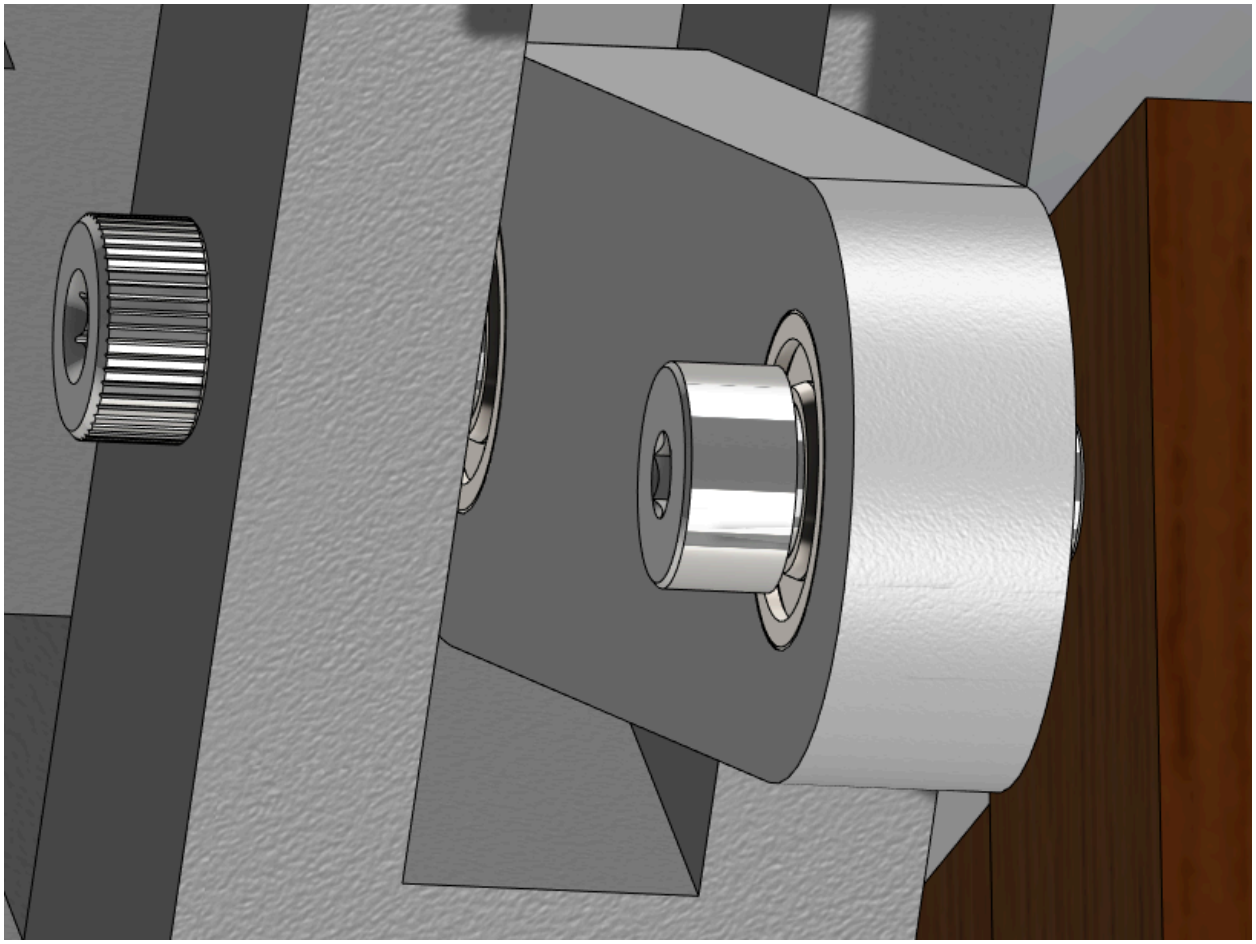


Figure 17. Upper fixed joint close up

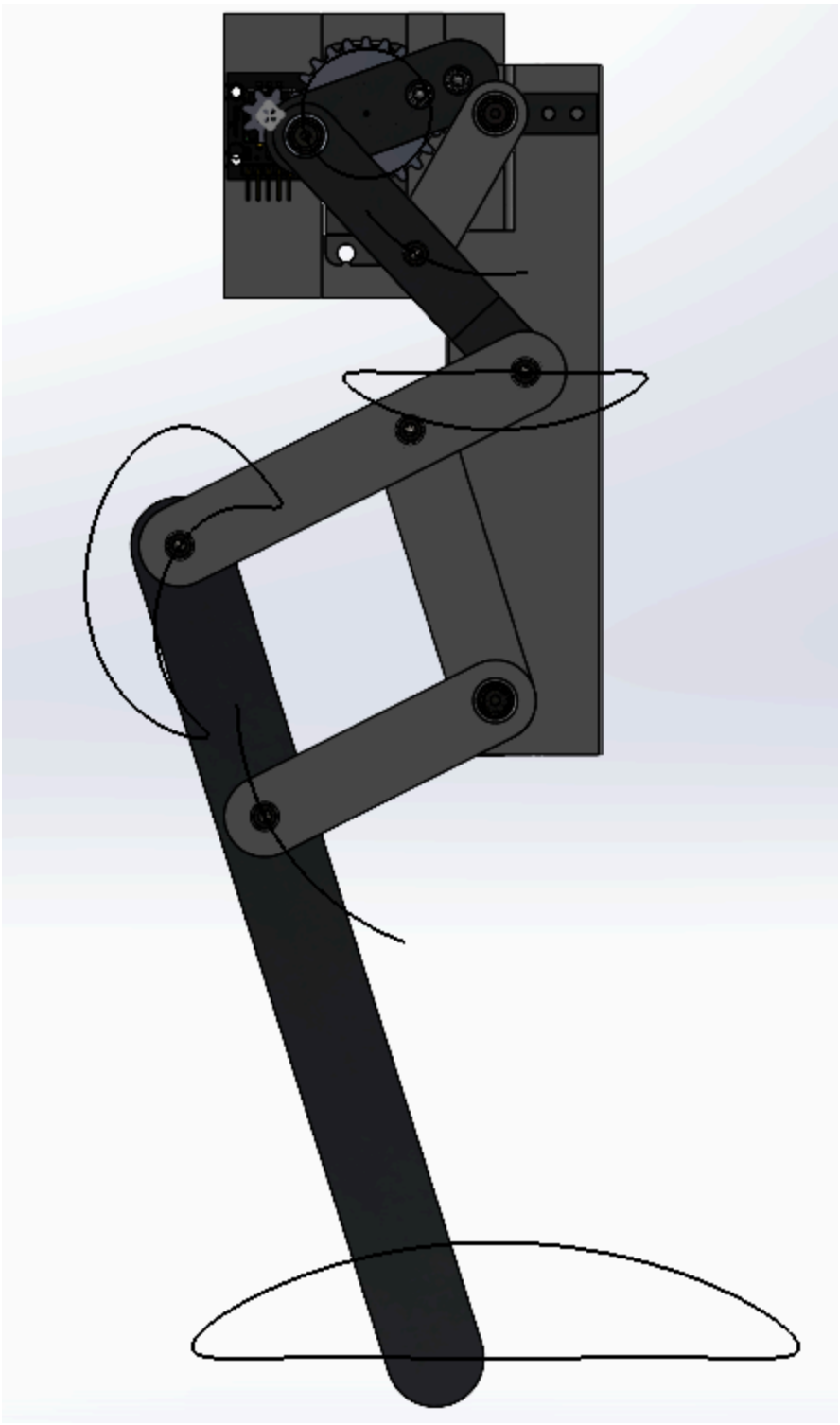


Figure 18. Final Leg Design and motion path

Now that we have established why the joint configuration is how it is, we established what links are outside or inside. The aforementioned link EDB which is connected to the crank is

also connected to the links CD and link BGH. Link EDB is an outside link at joint D and an inside link at joint B. This was determined to be the best configuration because this link transfers all of the force from the 4 bar mechanism to the pantograph, while holding some of the weight of the robot. The outside connection at joint D allows the link to evenly transfer the forces from the ground to the upper fixed via link CD. Link EDB transfers to an inside link at joint B to transmit the power to the pantograph, where link BGH is only an outside link. At joints G and H, this link is connected to GM and HIA respectively as an outside link, the others being inside links. The forces from the ground push link HIA upwards, and the forces are transferred to links BGH and MI at joints H and I respectively. Link MI transfers these forces to the lower fixed link, and link BGH does the same via link GM.

This leg design satisfies our design criteria in many ways. Firstly, as seen in the final design motion path, the mechanism achieves the desired motion path and transmission angles that were set out in 2D. This motion path will allow us to sequence the legs such that at least 3 of the legs are in the flat portion of the motion path at all times. This translates to a stable walking motion for the final robot, fitting the criteria. This design also efficiently distributes stresses from the motor and ground, as seen in static and dynamic studies found in the appendix C. These simulations had boundary conditions greater than what we expect the leg to experience, that being one leg holding half of the weight. As shown, the deformation, stress and strain are at acceptable levels for this exaggerated force, meaning that it will suffice when practically applied.

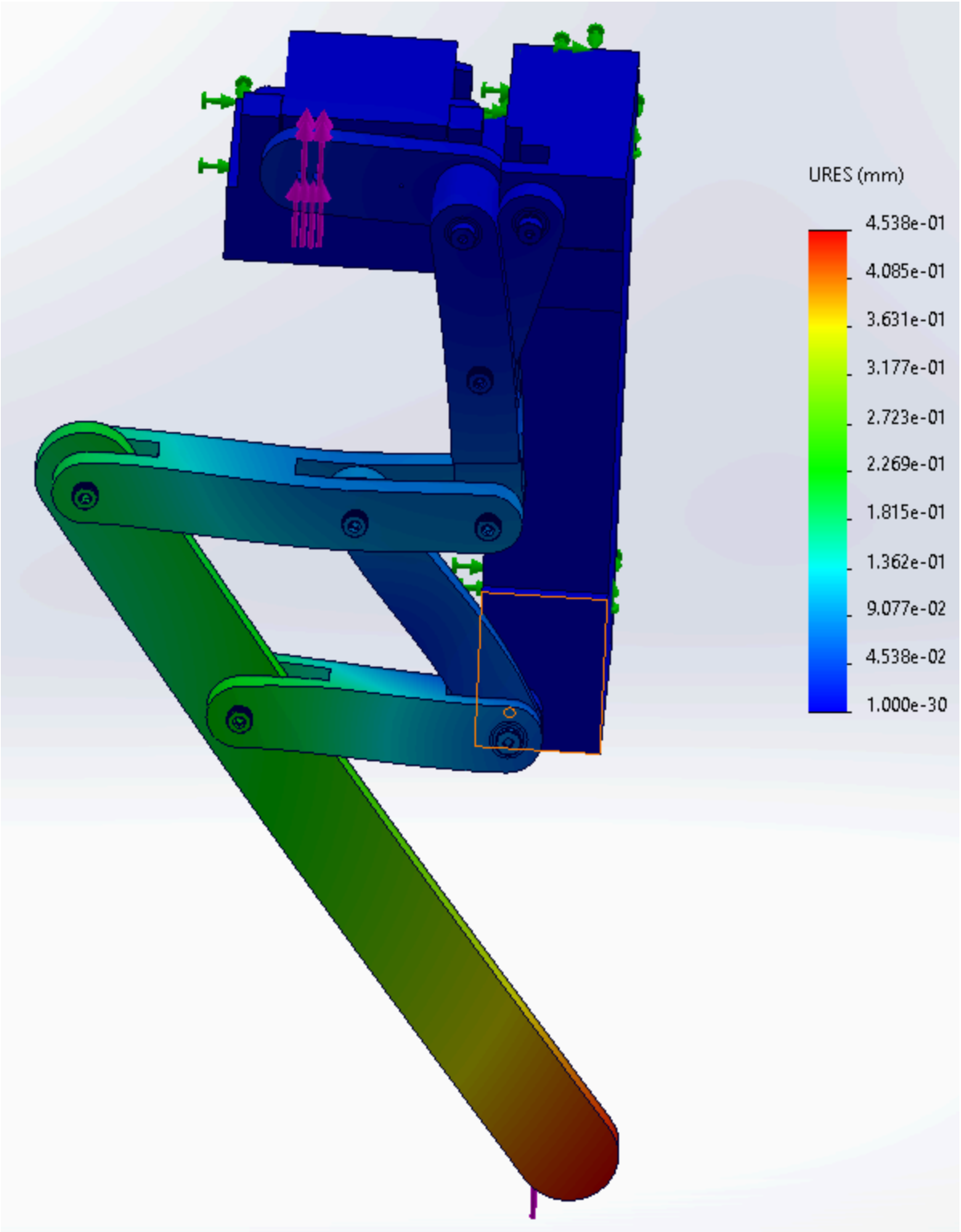


Figure 19. Static simulation with exaggerated forces

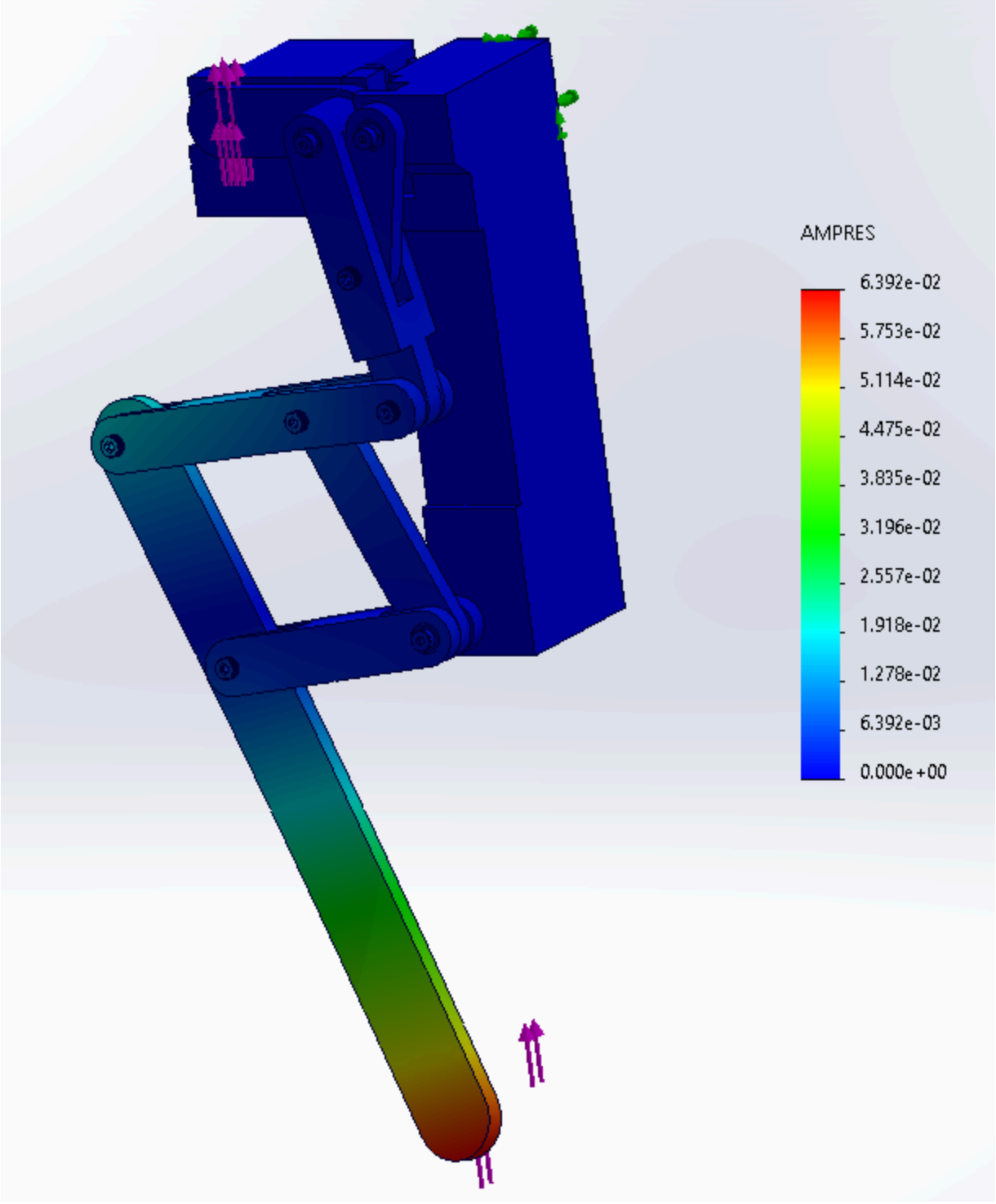


Figure 20. Buckling amplitude in mm

Electronics

Below is a basic flow chart for the electronic system of the walker:

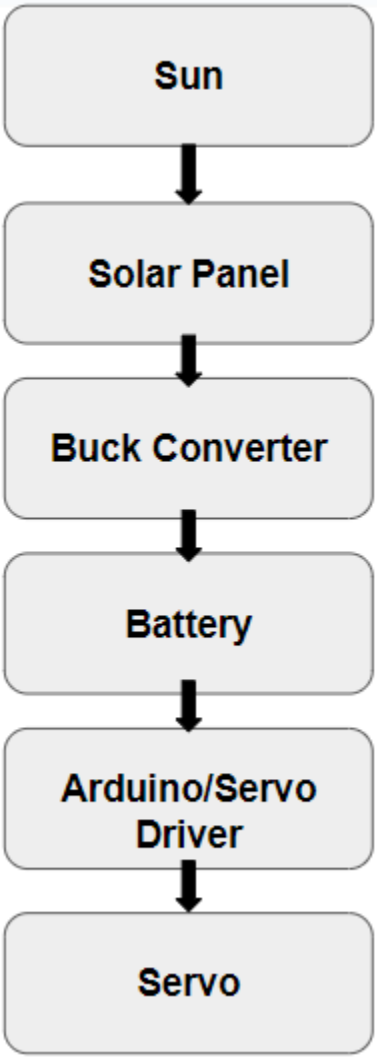


Figure 21. Electronics power flow diagram

We will be using a 20 Watt 12 volt solar panel to power our walker. It will be mounted directly on top of the chassis. The panel we are purchasing has a built-in controller to prevent overcharging of the battery. With the built in controller, we don't need to purchase an MPPT to control the voltage output of the panel. It has dimensions of 7.68"D x 14.37"W x 0.79"H and weighs 2.1 pounds. The dimensions of the chassis will be modified in order to fit the panel.

Connected to the solar panel will be a buck converter in order to step down the higher voltage output of the solar panel to 7.4 volts which is suitable for our selected battery. The buck converter ensures that the battery receives a safe and consistent voltage while charging.

For our battery, we selected a 5200 mAh 7.4 volt lithium polymer battery. Lithium polymer batteries typically have higher efficiencies than other battery types. They also have high energy densities which means they can store a lot of energy while being relatively light. This is beneficial for our walker because weight needs to be kept at a minimum to prevent failure of the legs.

An Arduino Mega combined with a servo motor driver will be used to control the motion of the walker. The motor drivers, lidar camera, and sensors will be connected to the board using the appropriate pins. We will write the code in Arduino IDE and upload it to the Mega, allowing the walker to execute any specified commands.

We decided to go with servo motors as our motor type. These motors have high torque and allow for precise control of position. This is essential for robotics where fine adjustments need to be made in the robots position. The motors we selected can provide upwards of 3.0 Nm of stall torque at 7.4 volts input. One motor will be connected to the crank in each leg using the servo horn and screws. The fifth motor will be connected to the rack and pinion to allow the walker to turn. The motor should be suitable for this application given the weight of the walker and size of the legs (see Appendix C for calculations).

Chassis Design

Once we decided that our turning mechanism would be like a car we set out to outline the design. Our turning requirements are for the front legs to turn and keep the chassis stable. The chassis has other functions in our robot. It needs to withstand the stress and forces when walking. It also needs to house all electrical components. Another requirement of the chassis is to be lightweight. The less weight the chassis is the less our legs will need to push. The less our legs need to push then, we can use cheaper and lower power motors. The chassis being overcomplicated can hurt the legs subgroup in their challenges. But we need to balance a lightweight chassis with strong material and effective turning machinery. But even more than light weight we want our chassis to be balanced. Our robot can not walk very far if it falls over. In summary, we need our chassis to be lightweight, durable, balanced, and capable of effective turning.

The two challenges of lightweight and durable can be tackled at the same time. This is through material selection. Once we have a basic design of the chassis dimensions we can begin simulation testing and research. I listed our two objectives but there is a sneaky third objective. We also want low cost. Our budget is only $700 if you remember. We can spend all our money on a carbon fiber chassis but then we won't have any money for legs and electronics. There is a common group of materials that fall under the category of low cost and lightweight. This group is called plastics. The focus of our group fell on ABS versus PVC vs PETG. These are two common plastics with great strength. We noticed that there were other plastics such as PLA that were cheaper however just by looking at material properties we knew this would not fit our project. There are many properties that we can look into to see how plastics can help us design our chassis.

When talking about strength of a material it is important to talk about what kind of stress or force is occurring. The stresses will be caused by the legs and the weight of the electronics. At a standing position weight is distributed even throughout the chassis. However when we are

walking there will be 2 points of contact per cycle. This means that we need to design the chassis so that it can withstand the forces from electronics weight and the normal force of 2 legs. The stress we will experience is shear stress. The property we must look at to understand shear stress is the Young's Modulus. Young's Modulus is based on the stress strain relationship of a material. By looking at the Young's Modulus of our material options we can estimate which material would be best for withstanding shear stress of the chassis.

Our two main options for plastic were ABS and PVC. These options are strong enough and relatively cheap. We can look up the Young's Modulus online to compare their strength. ABS and PVC have a Young's Modulus of about 4.2 GPa and 3.5 GPa. This shows us that ABS is stronger than PVC. So for durability we can say that ABS plastics is the stronger material. But if we look at cost PVC is slightly cheaper than ABS plastic. So this gives us a not so clear choice on what material is better. We anticipate based on our electronics to have to support a load of about 15 lbs. A load of 15 lbs is relatively light so we should be able to use the cheaper PVC to meet our requirements for the chassis. By using engineering properties such as Young's Modulus we can determine that PVC meets our engineering requirements and is the cost effective option.

Through discussion of the chassis design we realized that there is another group of materials that falls under low weight and cheap. This material is wood. While wood does not allow for as much customization as plastic it gives us more strength at a lower cost. Through minor changes in our design we were able to have a chassis design that was made out of wood. This would require us to get more components to assemble a chassis but overall would improve the design. Just from our experience working with engineering we were certain that wood would be a better choice over PVC plastic. The next step would be to verify our initial design through simulation and research. Although we selected PETG for our chassis material, we initially decided to use oak wood to investigate if our chassis design would be strong enough.

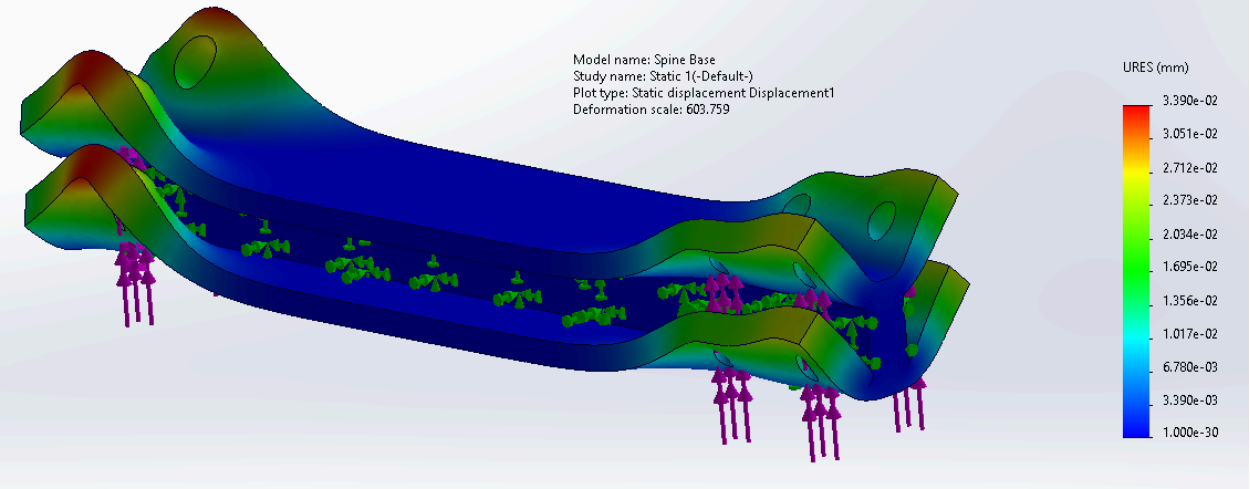


Figure 22. ANSYS Simulation of Chassis, Deformation Color Map

Since we know the weight on our chassis, we can see how it reacts to stresses. Our goal is not just to have a strong chassis but also have it be light weight. While we may have selected a strong material, there is much in the design of the chassis that can contribute to making it as lightweight as possible. We used a simulation software known as ANSYS to test if our chassis could withstand the stress we put it under. From each test we can simplify our design to make it more lightweight. Through a process of experimentation and changing certain variables we can

use the simulation to perfect our design. The results are shown above from a design that we feel is the most simple while still meeting stress requirements. Under the most extreme scenarios our chassis would experience deformation in millimeters. And this deformation is well within the elastic range of our material. This design removes extra barriers we had for strength to get the lightest possible design. By using software tools like ANSYS we were able to verify our design and perfect it to be as lightweight and strong as possible.

Now that we have a structurally strong chassis we need to discuss turning. Our design we tested uses 2 walls on a hinge that rotate to fit the angle needed to turn. But before I explain how our mechanism works it is important to understand how a car turns. Then I will distinguish how our design differs from a car and what challenges we needed to overcome. Finally, I will walk through the process where we designed our steering mechanism. The analysis of car steering and our adaptations will help us create a better steering system.

Car steering works by having all the wheels on the same turning radius. In order to do this a car must have its two front wheels pointed at different angles. If the wheels are pointed at different angles the following happens.

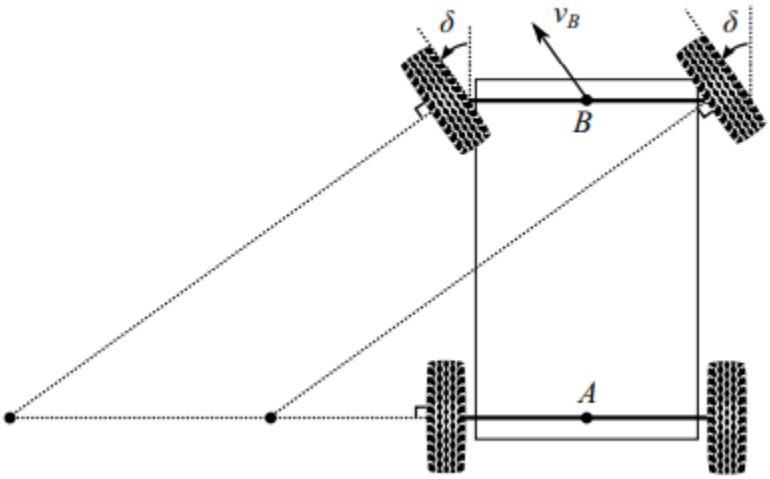


Figure 23. Improper Car Steering Diagram

A design like this would give the driver no control over turning. Even worse, the driver could get into an accident. The wheels must have the same instant center. There is a difference between the two front wheels. They actually run at different speeds. The distance the outer tire must travel is larger than the inner tire. So the outer tire must travel at a faster speed. If this did not occur it would cause the tires to slip resulting in a smaller turning. It would also feel very rough for the driver and it may cause them to lose control. So the front wheels of the car must have different speeds and angles in order to turn.

Car manufacturers found an easy way to make the tires different angles with the same motion change. If the tires start angled out slightly then when they are turned they will have the same instant center. This concept is somewhat confusing so there is a photo attached that explains this concept.

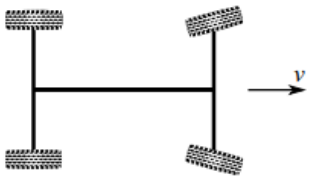


Figure 24. Car Design with Toe Out Tires

This works in cars with 2 wheel drive since the back wheel can be spun to match whatever the front tires need them to turn at. Cars with 4 wheel drive have a more complicated design for turning and changing the speeds of each tire. If we know what angle we want to turn at we can angle our wheels and achieve turning.

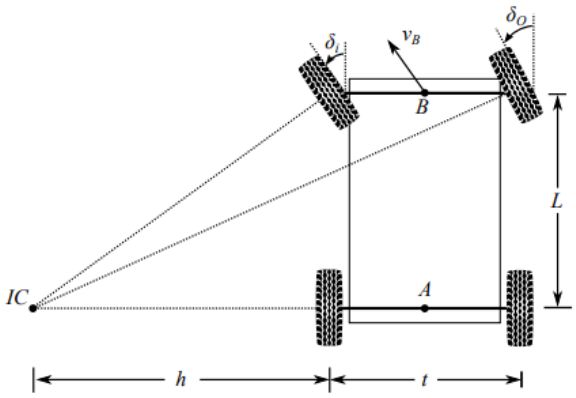


Figure 25. Fundamental Steer Equation

$$\tan\delta_i = \frac{L}{h} \qquad \tan\delta_o = \frac{L}{h+t}$$ **[29, 30]**

Above are the fundamental equations we can use to calculate the steer angles. If you look carefully at the equation you will notice that they are just basic trigonometry. Since we have the chassis dimensions all we need to do is decide our turning radius. Once we have that we can find both steer angles. The "L" or wheelbase of our chassis is 16 inches. The "t" or track is 12 inches. The h is a variable that is part of the turning radius of the inner wheels. This is something we can change depending on how sharp we want our robot to be able to turn. This may be something we decide to change through experimentation. For design purposes we choose 20 inches. Our robot must navigate our obstacle but it must also not turn to sharply causing it to lose balance. This value of h should satisfy both of these conditions. The following is a matlab code that does all the calculations for us. Then it gives us what we need the offset to be to achieve these angles.

```
%Intial conditions

%independant var
L = 16; %Length in
W = 12; %Width in

%Experiemntal var
H = 20; %Distance to IC, in

%Dependant
steerO = 0;
steerI = 0;

steerI = acotd( H/L);
steerO = atand( L / (H+W) );

disp("Offset Angle")
angleOffset = (steerI - steerO)/2
```

```
>> SteeringAngles
Offset Angle

angleOffset =

    6.0474
```

Figure 26. Coding to Solve the Steer Equations

Here we can see that we must offset the front by 6 degrees to achieve a turn with a radius of 20 inches. This means that the "wheel" would be turned by the same amount and have the same instant center but have a different angle. So using the fundamental vehicle equations we can solve for how much we need to adjust our legs in order to turn.

There were many iterations of ways to achieve this turning motion. The original design had two motors that turned the inner and outer "wheels" separately. Then we tried reducing the system down to one motor with additional mechanisms. This involved belt pulleys or a rack and pinion system. Both of these systems were good but far too complicated. Based on an idea suggested by Dr. Basily, we used a 4 bar mechanism with 2 fixed points.

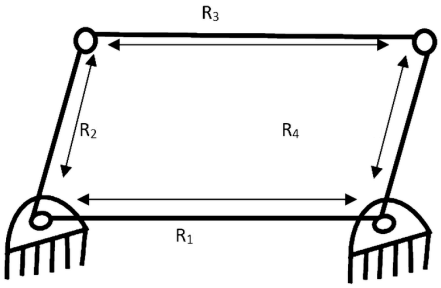


Figure 27. 4 Bar mechanism with 2 fixed points

The "wheels" would be attached to the links R2 and R4. Then the link R3 would move left and right to turn the two other links. This is similar to a rack and pinion but instead of using gears to move link R3 we are using a motor with a link that is attached to the R3 with a slider. This way will only move the link left or right. This design is extremely simple and cost effective. With this 4 bar design we can achieve the angles needed for turning.

Now I said before how we need to adapt our design. This is why we have been putting "wheels" when writing. It is because we have legs. The issue with legs is that we can not change the speed they move. If we change the speed they rotate at we will lose the walk cycle that keeps

us balanced. So how do we turn our legs if we can't change the speed? We just can. But in order to explain how we need to understand how slip works in a car. If we understand slip we can understand how to adapt our design.

Slip in a car tires happens when the front or rear tires lose traction. This causes a slip angle. The slip angle changes the steering angle. Front slip causes the car to understeer whereas rear slip causes the tires to oversteer. This concept is relevant to us. Another concept that is relevant to us is tank tread steering. Tanks do not change angles they only change the velocity of the left compared to the right side. In the same way our robot changes direction without changing velocity. So it is the opposite of tank steering. If we think about the robot as having a diagonal walk pattern we can break down the component force turning the robot. This is for a right turn. The back has a 100% positive y-direction component. The front has 90% positive y-direction component and a 10% x-direction component. This means that the same effect in Tank treads will happen with us. That means there is a moment that occurs turning the robot slightly left. This is in the same way a slip angle makes a car have understeer. This means the result is just a smaller turn but turning non the less. This may be something we need to experiment with but the principle is solid. So by using car and tank concepts we can explain how we can still turn even without changing the speeds.

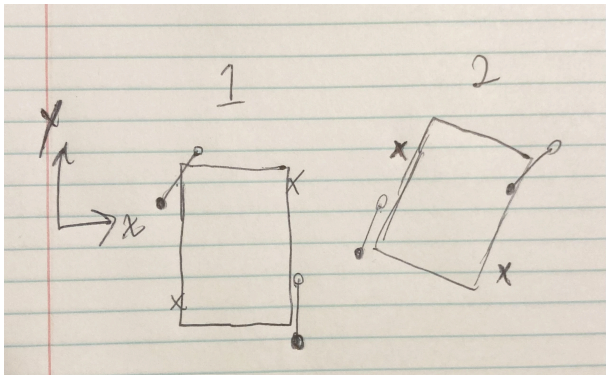


Figure 28. Rover Walking with Slip Concept Diagram

The chassis is a key part of our design. Our chassis must achieve turning for our legs. It also must balance lightweight design with structural strength. Using tools such as Matlab and ANSYS we can adjust our design. In order to use these tools we needed to understand the fundamentals of engineering concepts. These concepts range from material properties to vehicle dynamics. This all in all led us to a PETG chassis that uses a 4-bar mechanism to achieve steering angles. And using our knowledge of steering systems we adapted our design to meet the constraints of constant speeds required for our walking motion. The whole of our engineering experience allowed us to analyze our design and create a chassis that meets our goals.

4.1 MANUFACTURING PROCESS

Parts Choice Justification:

Motors:

Based on the estimated weight of 20 pounds and leg length of approximately 12 inches, we determined that a motor with 3 Nm of torque should be sufficient to move our walker. Servo motors provide high torque and precise control which makes it an excellent option for this application. They are cheap in comparison to regular brushless DC motors and come with a horn and screws to assist in mounting the motor to the leg module. We made a mistake in buying continuous rotation servo motors without encoders, so we bought encoders and mounted them next to the motors. We 3D printed gears with a one to three gear ratio to get a 6 degree precision in the motors rotation, allowing us to accurately control the motors.

Battery:

Lithium polymer batteries have high energy density which makes them a great choice in RC vehicles and robotics. We selected a 5200 mAh 7.4 volt 80C LiPo battery to store power from the solar panel. It should provide enough power to allow our walker to complete the race. Given that all 5 motors are operating at stall torque (which isn't going to be the case), at full charge, the battery should last around 20 minutes.

Motor Controller:

The motor controller will be used to control the servos. It is important to have motor controllers to prevent any damage to the Arduino caused by voltage spikes or excessive current draw. The motor driver will ensure that the performance of the motors is optimized through fine motor control and minimal energy losses.

Arduino:

We selected an Arduino board over a Raspberry Pi because we are more familiar with the Arduino software. In general, it is better to use Raspberry Pi boards for robotics due to it having more processing power. For our application, the Arduino Mega should be able to handle all of the electronic components.

Solar Panel:

We went with a 20 Watt 12 volt solar panel to supply power to our walker. The previous walker used two 5 watt 12 volt panels which isn't much power. According to the calculations, it took over 8 hours to charge the battery. With the 20 Watt solar panel, we cut that charge time nearly in half.

Lidar Camera:

Since the walker has to be autonomous, it needs some way of detecting any obstacles that might be in its way. Lidar cameras provide accurate spatial information by measuring the time it takes for light to travel to an object and return to the sensor. The lidar camera we plan on using is compatible with Arduino so we can tell the walker how to respond to obstacles it may face.

Chassis:

The chassis is made of 3D printed PETG H - bars that were simply connected together by mending plates and screws, washers, and bolts, similar to how I beams are secured in buildings. The holes for the screws were printed in the H beams to avoid imprecise drilling, and the screws were amply torqued down. The chassis is composed of 12 of these H beams making it 16 inches long and about 8.25 inches wide. The H beams in the front are attached to flat plates, also 3d printed, by L brackets, similarly secured by screws. The flat plate in the front is to allow the easy attachment of the pivots joints, which the leg bases of the front legs are then attached to to allow for turning. The back H beams are longer and are directly connected to the leg base of the back legs. They are longer such that the space taken up from the pivots is accounted for, so the front and rear legs are in line with each other along the "spine" of the chassis. Including the width of the leg bases, the width of the total robot is 12..15" and 16" long. We decided to use H beams because it reduces weight significantly, while still being able to take the loads from the robot. We ran FEM models on one of the beams with 40 lbf on both ends, an extremely exaggerated force that the beam sufficiently withheld well below the yield stress.

Metal Links:

The leg mechanism has 7 total physical linkages, three of which are made of aluminum. The three that are made of aluminum are the crank link HIA, or the link that makes contact with the ground, and link EDB. Originally, all the links were going to be made out of 3D printed abs plastic, however we discovered that would be impractical. After running both static and dynamic simulations of the leg link, we discovered that the desired geometry of link HIA would deform plastically when the material was abs. To mitigate the deformation, we enlarged the link and found that it could not practically be enlarged enough, so the change to aluminum was finalized. We also found that link EBD would plastically deform, however it could not be enlarged practically. The switch to aluminum for these links avoids plastic deformation and gets a higher factor of safety.

Similarly, through simulations we found plastic deformation of the crank would occur if it was made of abs. The geometry of the crank could not be enlarged due to its close proximity to the upper fixed joint, so replacing the abs material was essential. Considering the small size of the crank, the weight and cost increase with going with aluminum was small, so the choice was obvious. The increases were easily offset by the increased safety factor and strength that comes with the aluminum.

Plastic Link:

The remaining 4 of the 7 links are going to be made of 3d printed abs plastic. The plastic is not as strong as the aluminum, however the larger geometry and orientation of these links do not require the extra strength aluminum provides. To use aluminum for these links would increase the cost and weight of each leg unit, while providing little benefit. This is even more true with our new design. Our old design of these 4 links had 9mm holes for our 9mm outer diameter ball bearings for all joints, however those have been replaced with a 4mm diameter and threaded M3 holes for the shoulder screws for the outside link. This new configuration has smaller holes when not using the ball bearings, meaning the stress concentration from these holes will not be as severe. Since the 9mm holes were not a problem with the plastic links, we knew that it would not be a problem with 4 and 2.5 mm holes for the shoulder screws, which was confirmed with simulations.

Joint configuration(bearings and shoulder screws):

The aforementioned shoulder screws will be used to secure the joints in 4 configurations, the crank joint, upper fixed joint, lower fixed joint and the moving joints. The ball bearings we used are deep groove ball bearings with an inner diameter of 4mm, outer diameter of 9mm, and a 4mm thickness. The deep groove ball bearings are excellent at distributing radial load, which is the predominant force, and is also good at thrust, or axial loads. This recommendation was from Dr. Basily, who affirmed that deep groove bearings were the best fit for our application. Dr. Basily also recommended the shoulder screw because when they are secured by nuts, the smaller thread size prevents the nut from touching the outer ring of the ball bearing it is securing, therefore preventing friction. Small washers will also be used between bearings or the link walls, depending on the joint configuration. These washers prevent movement along the axis of the shoulder screw/joint, preventing the links from rubbing against each other and getting stuck at weird angles. The shoulder screws will be secured by 2 set screws to prevent the shoulder from rotating and traversing along its axis, further securing it beyond what the threads already do.

The Crank joint, as mentioned in the design portion uses 3 ball bearings, one in the crank and 2 in link EDB. This is the only joint that requires a nut to secure it, which replaces the set screws and the threaded M3 hole. The upper and lower fixed joints have the shoulder screw simply inserted into the bearing of the respective links with washers spacing them from each other and the base. The shoulder screw is then screwed into the M3 holes made for them. Similarly, the moving joints are inserted into the 4mm hole on one side of the outside link, then into the bearing of the inside link, then screwed into the other side of the outside link. The set screws are then used above and below the 4mm outside link's hole, securing the shoulder screw.

Leg Base:

Due to the aforementioned geometry of the leg base beam, we decided that 3d printed PETG H beams would be the best material to use. Originally it was going to be oak such that the motors stand and support would be one rigid body. We found that the plastic is denser and weaker than oak wood, however it has no chance of warping and is easier to tap and thread than the wood. With this the choice to go with PETG was made, in combination with making the chassis PETG. This allows for easy connection to the chassis via pivot points or brackets. The choice to go with wood also helped with the ease of manufacturing. The base requires a weird geometry, so the motor mount and a spacer for the upper joint were printed separately, and screwed down to the H beam leg base.

4.2 Manufacturing and Assembly

Metal Machining

We machined 3 different parts for each leg, totaling 12 from 6-61 aluminum. First was the crank where we first milled it down to a rectangular size, then while still zeroed we drilled 2 holes to attach to the servo horn. The holes were threaded and another smaller hole was drilled for the screw's head to sit inside. Then, another hole was drilled for the bearing, and we then used a 0.354'' reamer to achieve the desired interference, which is discussed later in the standards section. The edges were then rounded to reduce weight. Next was the link EBD, another high stress component which connected to the crank. Similarly, we milled it down to size, then drilled the holes for the shoulder screws, then drilled and reamed the holes for the

bearings. We then had to make a slit for a link to fit into, and we achieved this by drilling a series of holes, then flattening the slit via milling. Again the edges were rounded. Lastly was AIH, the bottom link which touches the ground. Like the others it was first milled to size and the holes for the bearings were drilled and reamed. Like the others, the edges were rounded, however we made a mistake in measurements and the initial bearing holes were slightly off where they were supposed to be. To remedy this, we simply drilled and reamed new holes, and left a bearing in the hole that was off to reduce the stress concentration from the hole.

PETG 3D printing

Many of our parts discussed in the previous section were 3D printed out of PETG plastic. We first printed a few links and spacers to test out the design, and upon confirmation we printed all 4 plastic links of the leg mechanism at 70 percent infill. We then printed the H beams for the leg base, then the H beams composing the chassis, also at 70 percent infill. These components experienced relatively high stresses, so the high infill was certainly justified. We found that there are diminishing returns on infill percentages above about 60 percent, so we did not go to 100 percent to reduce weight and printing time. Lastly, the motor/ encoder mounts and the turning mechanism were also printed, however only at 15 percent infill. These components do not experience much stress, so to reduce weight, we printed them at low infill.

Leg Mechanism Assembly

We began the leg assembly off of the leg base, where we used the 1 and 2 mm spacers on the shoulder screws to avoid friction between the links at the joints. Once all 7 links were together, we attached the upper spacer for the upper fixed joint to the leg base, then threaded the shoulder screws into the respective upper and lower fixed joints, using the spacers to avoid the mechanism from rubbing on the leg base. The motor mount was then attached to the leg base, and the servo motor was attached to the crank via the servo horn. This resulted in the final leg assembly, where there is very little wiggle in the leg due to the spacers, and very little losses due to friction.

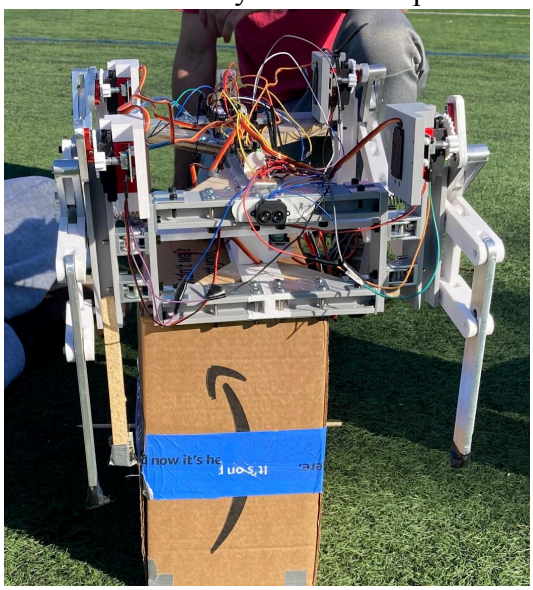
Chassis and final Assembly

In the 3D printed H bars for the chassis, we printed M5 holes separated such that we could fasten mending plates to secure the H bars together. We also included a vertical beam in the middle of the robot to minimize bending and ensure the desired separation of the upper and lower halves of the chassis. L brackets to secure the leg bases to the chassis were first attached to the leg bases themselves, then screwed down into the chassis. At the front, a plastic plate at each side was attached to the chassis, which in turn was attached to 2 pivot joints directly inline with one another vertically. The leg base in the front was then attached to the pivot joints, which allows for the turning of the front legs. The pin attachments for the turning mechanism were then attached to the leg bases in the front, and the motor mount for the turning mechanism was placed. The pivot arm was attached to each of the pin attachments, then attached to the motor, thus completing the turning mechanism.

Electronics

Our electronics system is composed of an Arduino Mega, a servo drive, a lidar camera, a 5200mah battery, a 9 volt battery, bluetooth module, and a solar panel. We ran a wire from the ground and a 5 volt source from the Arduino pins and put them on a separate solderable board and soldered it on. We then powered the encoders, servo drive, bluetooth module, and lidar by soldering them to their respective positions on the solderable board. We then connected all of the info wires to their respective pins on the Arduino in accordance with our code. The 5200mah battery was soldered in with the servo driver, and the servo motors were connected to their

respective position on the drive. We used a 9 volt battery to power the Arduino, and the bluetooth module to remotely start and stop the autonomous code on the Arduino.



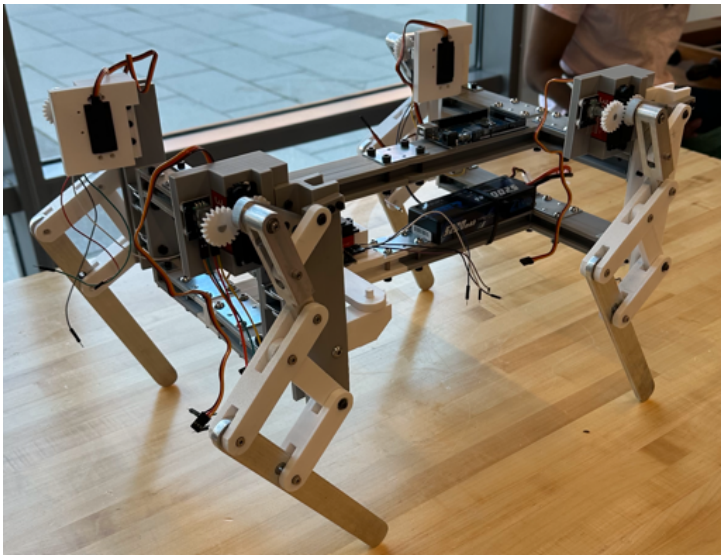


Figure 29a and 29b. Final prototype of SolVoyager

5.1 TESTING

Once the design was finalized, a single leg unit was assembled using entirely 3D printed materials. During this time the components of all four legs were either in queue or were in the process of being manufactured. The single leg prototype was attached to a stationary testing rig to confirm that it operated as desired. Clearances between linkages, joints, and bolts were noted and incorporated into the design and spacers were created to prevent the components from striking one another during motion.



Fig 30. Prototype leg attached to vertical test rig

Electrical components were each tested separately with every subsystem receiving a thorough check. The solar panel and battery were attached to one another using gator clips and the charge was monitored to confirm both parts were functioning and compatible with one another.

The solar panel was tested to determine the time it will take to charge the battery. Below is a summary of the results from testing:

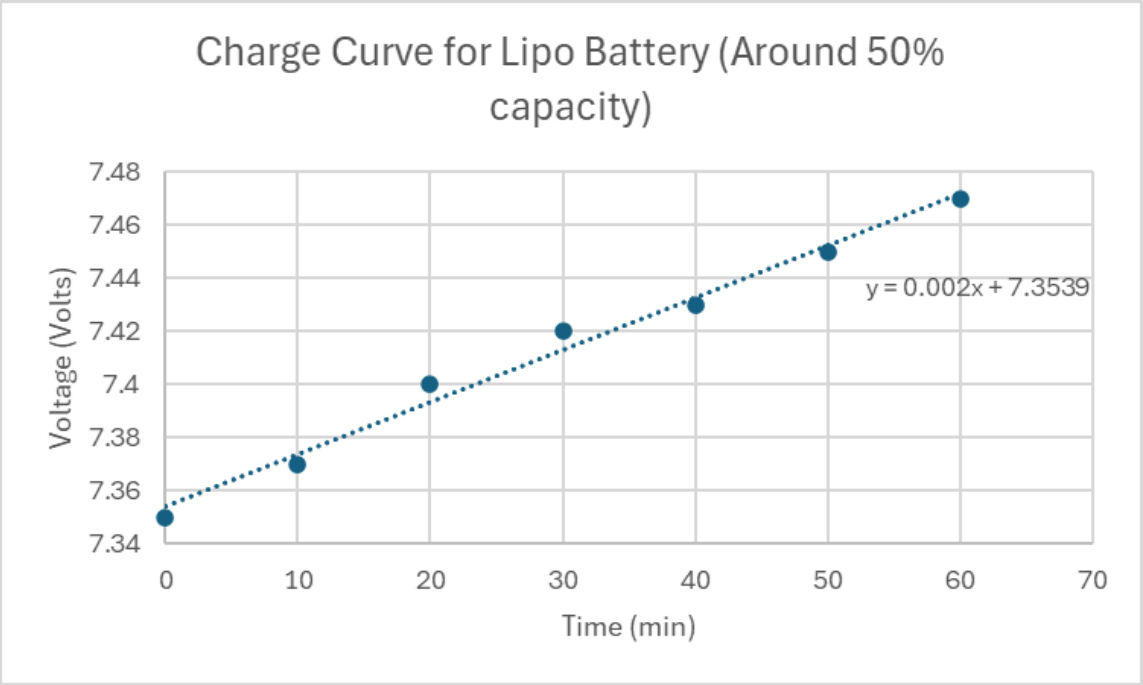


Figure 31. Voltage vs. time curve for the solar panel

- Measurements taken March 14th, 4:45 to 5:45 PM, Piscataway, mostly sunny, some clouds
- Solar panel was positioned roughly normal to the sun
- Open circuit voltage of 18 volts (matches the given value from manufacturer)

Assuming 0% charge is 3.0 volts/cell and full charge is 4.2 volts/cell, it would take 8.3 hours to completely charge the battery. This is similar to the theoretical charge time of 13.94 hours obtained in Appendix C.

All five motors were attached to a simple circuit with the Arduino to confirm that they were functional. Once confirmed, a leg motor was attached to the aforementioned leg test rig and an Arduino. During the whole process wires were strictly color coded and monitored in case of failure. Mechanical and electrical components were kept apart by the test rig to prevent any entanglement or shorting.

Chassis and leg components were routinely assembled as each part was acquired or manufactured to test fit each piece. Because not all components were produced at once, each piece was tested separately for fitting. The chassis parts were tested for any warping by carefully measuring them with a straight edge and calipers. During test fitting any clearance issues with other components, links, or bolts were noted for modification. There were some discrepancies with the parts, specifically with clearance between bolts in some areas. Where necessary bolts were either shaved or chassis components were modified to accommodate the bolts.

5.2 Standards

Steven Kaplan: ANSI C18.2M (Portable rechargeable batteries specifications)

This standard ensures that all of the specifications of the battery we used match the values given by the manufacturer. The dimensions of the battery we obtained were very similar

with the battery measuring 5.40” x 1.85” x 1.00”. The voltage and capacity were determined using a lithium polymer battery charger. We determined that the maximum voltage the battery can handle is 8.4 volts total. The 7.4 volts advertised is the nominal or average voltage of the battery. The capacity of the battery was determined to be close to the given 5200 mAh value.

Brendan Curran: Bearing Standards: National Testing Network Bearing Standards and ISO 286-1

We used the National Testing Networks Bearing standards to find out what type of interference we needed for our conditions. Inherent with the mechanism's geometry, the load on the bearings is an unbalanced load, and only the outside ring of the bearing, the one that touches the linkages, would be rotating. The NTN recommends in this case a tight fit on the outside ring for the best radial force transfer, and a looser fit on the inside ring.

Table 7.1 Radial load and bearing fit

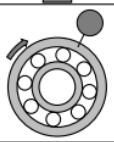
Bearing rotation and load	Illustration	Ring load	Fit
Inner ring : Rotating Outer ring : Stationary Load direction : Constant	 Static load	Rotating inner ring load	Inner ring : Tight fit
Inner ring : Stationary Outer ring : Rotating Load direction : Rotates with outer ring	 Unbalanced load	Static outer ring load	Outer ring : Loose fit
Inner ring : Stationary Outer ring : Rotating Load direction : Constant	 Static load	Static inner ring load	Inner ring : Loose fit
Inner ring : Rotating Outer ring : Stationary Load direction : Rotates with outer ring	 Unbalanced load	Rotating outer ring load	Outer ring : Tight fit

Figure 32 : National Testing Networks Bearing standards fit recommendations

The looser fit on the inside ring was achieved with the shoulder screws, where the inside ring of the bearing was about 4.02mm in diameter and the shoulder screw was 3.98mm in diameter, giving an interference of .04mm, a loose fit for the small size of the diameters. The outside ring of the bearings fit was achieved differently for the PETG and aluminum links. For the PETG, we simply printed 9mm holes, and press fit in the bearings which have an outside diameter of 8.98mm. As Dr. Basily recommended and predicted, with the high elastic modulus of the PETG, it deformed and resulted in a very tight fit for the bearings, which upon testing was very difficult to remove. This was different from the aluminum as we had to drill the holes ourselves, for which the shop was in mostly imperial measurements which we used. First, we drilled the holes to be slightly smaller than the desired, then used a 0.354” reamer to get the desired interference. This resulted in an interference of 45/10000”, which was slightly more than the 35/10000” we wanted. We knew it was practically impossible to achieve the desired tolerances, so we bought green loctite 609 retaining adhesive to make up the difference. This resulted in the same as the PETG links, where the bearings had a tight fit which was very difficult to remove.

Sean Sullivan: IEEE 1679.1-2017 (Guide for the Characterization and Evaluation of Lithium-Based Batteries in Stationary Applications)

This standard is from the Institute of Electrical and Electronics Engineers. It outlines standards manufacturers must follow for the manufacturing of batteries. This report specifically focuses on lithium batteries and their unique properties. This standard is important because it can help us learn about different types and choose the proper battery for our system. This can also help us to have a better understanding of maintenance of our batteries. Improper maintenance can lead to inefficiencies and even possible threat of battery fires. This is why understanding this standard is important to our project.

There are many small takeaways from this standard. The first takeaway is the unique properties of lithium batteries. Lithium batteries are best suited for lightweight and long term application. This is perfect for remote drone and rover use. The lightweight can allow our motors to carry less weight increasing overall efficiency. Other batteries may give more power but would have other negative effects with higher weight and short term use. Another keypoint the charger methods. Lithium batteries work best when they do not reach 0% or 100%. This means to extend battery life we should be aware of the current charge. There are also safety concerns when using improper charging methods. Not using the recommended charging techniques can lead to draining the battery life or even battery fires. We made sure to follow these guidelines when designing our system. The final piece was discussing general safety around lithium batteries. This gave us an understanding of what signs may indicate a safety violation. This includes dents in the battery or strange fluctuations in power. Overall, this standard is important for implementation and safety of lithium batteries in our project.

Kirat Gill: OSHA 1926.963 - Testing and test facilities

OSHA Standard 1926.963 outlines safety requirements in construction and electronics testing facilities. Much of the standard pertains to the resources an employer must provide to employees however there are a few very important standards within it. Proper safety precautions must be taken when using electronic components. Proper shielding must be maintained between electronic components and moving mechanical components to prevent entanglement and shorts.

This was done through the vertical test rig. The wooden post on which the test leg was mounted provided a degree of separation between the wires and moving parts. On the Solvoyager itself, a smaller plastic component holds the motor in place and prevents wires from falling into the area in which the legs move. Additionally, during both testing and in final assembly, each wire is color coded or labeled so that its purpose can be seen immediately.

Justin Bradley: OSHA Standard 1910.132 (Personal Protective Equipment)

OSHA Standard 1910.132 focuses on the general personal protective equipment requirements that must be met when potentially being exposed to a hazard. This standard was implemented in the machine shop where there are some PPE rules such as closed toed shoes and eye protection. Goggles are provided with alcohol wipes to sanitize them before and after each use. Oils are also used a lot in the machine shop so there is provided oil removing powder in the bathroom. In the case someone wasn't dressed up to standard, they were asked to leave the shop.

Leo Shaw: OSHA Standard 1910.212 (General Requirements for all machines)

OSHA Standard 1910.212 outlines general requirements for machine guarding. Machine guarding consists of one or more methods to protect the operator and other employees in the

machine area from hazards created from point of operation, such as rotating parts, flying chips, or sparks. This standard was implemented for the milling machine which was used to manufacture some of the leg linkages. To verify that this standard was followed, we checked to see if the milling machine guard and safety shield were active and if the machine was properly anchored. Both of these conditions were satisfied which means anyone operating this machine should be safe from the rotating parts of the machine.

6. CONCLUSION

Throughout the semester, we designed a solar powered terrain walker we call SolVoyager. It has 4 legs which each utilize a Chebyshev mechanism and a pantograph to walk. Each leg has one servo motor to achieve a walking motion. The steering mechanism is similar to that of a car where the front legs can pivot in order to turn. One servo motor will be used per leg to assist in this motion. A solar panel will be placed on top of the walker which will feed energy to our battery to power the motors. SolVoyager will also have room in the chassis to carry a small 8 ounce payload. We just placed an order for electronics and fasteners and will have the leg linkages 3D printed and manufactured shortly.

In the beginning of the Spring semester, we started by testing the leg design to determine if it is able to achieve a desirable motion path. We successfully constructed one leg mounted to a stationary board. We tested our electronics to make sure they work and are suitable for this project. This involved verification of the solar panel and battery. We also assemble the motor and Arduino separate from mechanical components to verify the design before full assembly. Once we determined if the legs and electronics work properly, we began to assemble the entire robot. We spent a large amount of time coding and programming SolVoyager to be autonomous. We met in person as a team frequently to test the walker and make sure it's working as intended. These meetings involved fine tuning of coding and minor mechanical fixes. If everything goes to plan, we will have SolVoyager ready in late April for the race and Expo display.

SolVoyager was a success. Our goal was to construct a solar powered walking robot that can maneuver around obstacles and carry a small payload. We achieved this and innovated on what previous years have done. We implement a Chebyshev mechanism and a pantograph to walk. This reduces the motors required from 18 to only 5. This drastically improves efficiency and cost. We implement skills learned through our time at Rutgers to construct a solar power electronic system running on an Arduino. We also used many simulation tools to design mechanical components to meet certain stress contracts. Manufacturing and testing of SolVoyager required us to be adaptive to real world feedback of our designs. Given more time and a larger budget there are some improvements that we could have made to SolVoyager. Improvements in motor output could have enabled better performance. Due to our budget constraint we choose a lower cost motor that may have hindered higher possible speeds. There may also be more room for optimizing the programming used in the walking and turning cycles. There are certain delays that given more time could have been sorted out. SolVoyager achieved all its intended goals and did this by innovating compared to previous years.

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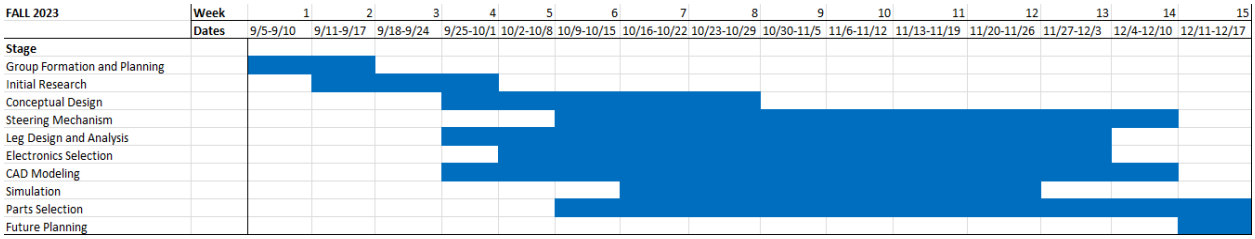
Appendix A. PROJECT MANAGEMENT

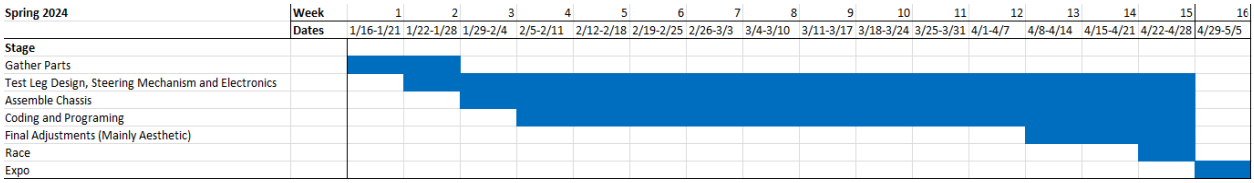
We used GroupMe and Discord to communicate with each other throughout the year. There were 1 or 2 internal team meetings every week to discuss progress, what our next steps were for the project, and work on TA reports or other assignments for the Senior Design class. It was difficult at times to schedule meetings where everyone can attend because of other classes and jobs. In the end, we were able to find times where all of us were free using a scheduling app. We started the Fall semester by researching different walking robot designs. We spent the first couple of weeks picking between a 6 leg or 4 leg robot. After discussing the pros and cons of each design, we decided as a team to go with the quadruped puppy robot design. After selecting the quadruped robot design, we split off into smaller sub teams to work on the subsystems of the walker. The project was divided into 3 main parts: leg design, chassis/steering mechanism, and electronics. There were several issues we faced with each of these subsystems. For the legs, there were issues with the kinematics and stability. For the chassis/steering mechanism, there were issues with the weight distribution of the single hinge design. For the electronics, it was difficult to determine a suitable motor since we weren't sure how much the walker was going to weigh. After the feasibility meeting with Dr. Basily, we had a good idea of how we wanted to tackle the project. We began to finalize the leg model, create a 4 bar steering mechanism, and select appropriate electronics from the insight given by Dr. Basily as well as our advisor and TA. We performed a finite element analysis on the leg system and determined where most of the stress was in the leg. The leg link in contact with the ground and the crank experienced the highest stress so we decided to use aluminum for these parts instead of PETG plastic. For the chassis, we decided to settle on PETG plastic as our material of choice due to its good mechanical properties and low cost. It will also be relatively easy to mount the leg module and other important components to the PETG chassis. After the manufacturability meeting with Dr. Basily, we began to finalize our parts list. Dr. Basily gave us valuable insight on the screws and fasteners that we should use to hold the leg linkages together, as well as some of the electronic components such as motors. We then finalized our parts list and put in the initial order.

For the Spring semester, we began by collecting the parts we ordered at the end of the Fall semester. All of the parts ordered were received and functioning. The team split into three subteams: electronics, machine shop/manufacturing, and design. The electronics subteam (Steven and Kirat) were responsible for testing the battery, solar panel, and motors to ensure they were suitable for the project. The main issue with the electronics was that the motors didn't have built in encoders. This was fixed by ordering an external encoder and 3D printing a gear that can be connected to another gear on the servo motor. This is essential for tracking the position of the motors. The machine shop/manufacturing subteam (Justin and Leo) were responsible for manufacturing the metal links as well as PETG parts for the chassis and leg module. The manufacturing process was very time consuming so we had to get started with that early on in the semester. The design team (Brendan and Sean) were responsible for updating the leg design and creating the chassis/turning mechanism. The leg module had to be updated so the motor could be properly mounted. The chassis/turing mechanism wasn't designed in the Fall semester due to time constraints so we completed that early in the Spring semester. The whole team came together for the assembly and coding/process. We ran into several issues with the assembly of the robot such as screws being hard to fasten and the parts intersecting with one another. With the tools we had, we were able to overcome these problems. The coding and programming was

mostly done by Justin and Leo. The biggest issue here was designing the PID controller for the robot. Deciding what the gain values should be for the controller was a very tedious process, but was essential for keeping the robot stable. Overall, the team worked well together to make SolVoyager a working product.

Figure A.1 and A.2: GANTT Chart for Fall 2023 and Spring 2024 semester:





Appendix B. PARTS LIST & BUDGET

Table B.1: Parts list and spending activities of the project.

Vendor	Part Name/Description	Part ASIN/ Part Number	Quantity	Unit Price	Total Price (Including tax and shipping)	
RUTGERS	Rutgers MAE Funding - Default Amount				$700	
AMAZON	QEBIDUM LM2596S DC-DC Buck Converter DIY 4-40V to 1.25-37V Step-Down Regulator Power Module Volt Stabilizer with LED Voltmeter Display	B07BTGRPQG	1	$6.99	$7.45	
AMAZON	Readytosky Digital 30KG Servo 360 Degree High Torque Metal Gear Servo Motor with 25T Servo Horn for RC Car Robot Arm Helicopter	B07Z3VGZNP	5	$23.88	$133.25	
AMAZON	MakerFocus Lidar Range Finder Sensor Module TF-Luna, Single-Point Micro Ranging Module 0.2 to 8m Compatible with Pixhawk, Arduino and Raspberry Pi with UART / I2C Communication Interface	B088NVX2L7	1	$23.99	$24.99	
AMAZON	**ELEGOO Mega R3 Project The Most Complete Ultimate Starter Kit with**	B01EWNUUUA	1	$65.99	$70.36	

	Tutorial Compatible with Arduino IDE					
AMAZON	uxcell 684ZZ Ball Bearing 4mm x 9mm x 4mm Double Shielded 684-2Z Deep Groove Bearings, Carbon Steel (Pack of 20)	**B07FW1958S**	**3**	**$9.49**	**$38.49**	
AMAZON	Zeee 2S Lipo Battery 5200mAh 7.4V 80C RC Hard Case Deans Plug for 1/8 1/10 RC Vehicles RC Car RC Truck Tank RC Buggy Truggy RC Airplane Racing Vehicles	B07MJ9H47D	**1**	**$19.58**	**$21.31**	
AMAZON	Command Picture Hanging Strips, Damage Free Hanging Picture Hangers, No Tools Wall Hanging Strips for Living Spaces, 12 Medium Pairs and 16 Large Pairs (56 Strips)	B073XR4X72	**1**	**$21.89**	**$23.34**	
AMAZON	RLECS 4PCS 2020 Zinc Alloy Movable Hinge, Die-Cast Zinc Alloy Pivot Joint, 27.5mm/1.08inch Aluminum Profile Accessories, Movable Hinge Silver	B0CF1X9R99	**1**	**$11.99**	**$12.78**	

AMAZON	GSCIT Club 12"x 8" Birch Plywood 8pcs, 1/8" Plywood Sheets Birch Plywood Wood for Crafts, Thin Wood 3mm Plywood for Laser Cutting and Engraving, CNC Cutting, Painting, Fretwork	B0CGZHJ48L	1	15.99	$17.05	
AMAZON	JEGONFRI High Hardness Wood Screws Assortment Kit, 209 pcs, Philips Drive Flat Head, Wood Screws, Screws, Drywall Screws, Assorted Screws, Screws Set, 6 Sizes (2”,1-1/2”,1-1/4”,1”,3/4”,1/2”)	B0BRPQW3G9	1	7.96	$8.49	
MCMASTER	Metric Alloy Steel Shoulder Screws	92981A745	4	3.2	$12.80	
MCMASTER	Metric 316 Stainless Steel Shoulder Screws	90269A209	4	10.15	$40.60	
MCMASTER	Metric Alloy Steel Shoulder Screws	**92981A142**	4	3.17	$12.68	
MCMASTER	Same-Size Thread 18-8 Stainless Steel Shoulder Screws	91273A368	20	4.32	$86.40	
AMAZON	ELEGOO Rapid	**B0CJ3295GB**	3	$23.79	$71.37	

	PETG Filament 1.75mm White & Gray 2KG, High Speed 30-600mm/s 3D Printer Filament Dimensional Accuracy +/- 0.02 mm, 1kg Cardboard Spool(2.2lbs) Fits for Most FDM 3D Printers					
AMAZON	2 Pcs 3/8" x 3" Aluminum Flat Bar 12" Long 6061 General Purpose Plate Extruded Aluminum Bar Solid Extruded Plate Mill Stock	**B0C6KDGK1S**	1	$24.99	$24.99	
	TOTAL QUANTITY		52	**TOTAL COST**	**$606.35**	

Most of the electronics were purchased in the initial parts order. Some parts are not included such as the solar panel and motor driver since they were blocked by the Rutgers Marketplace. For the second order, we mainly purchased 3D printer filament so we can do the 3D printing on our own. We have $93.65 remaining for the funding from the department.

Appendix C. DRAWINGS/ SIMULATION/CODES/DATA/DIAGRAMS

Below are calculations for the battery and torque required by the motor:

Energy Analysis:

Battery life: (5.2 Ah)/(5 motors*3.4 A) = **18.35 minutes minimum run time**

Charge time:

Specs: 7.4 Volts, 5.2Ah battery, 20 W solar panel

Following procedure from
https://voltaconsolar.com/blog/2021/04/27/how-long-does-it-take-to-charge-batteries-from-solar-panels/

We get a charge time of about 2.51 hours for our system

Taking into account efficiency of solar panel of 18% leads to charge time of **13.94 hours**

Rough Torque Analysis:

Assume the mass of walker is 15 lbs = 6.804 kg

Each leg will support 25% of the weight

Weight of walker = 6.804 kg*9.81 m/s^2 = 66.7 N

So 66.7/4 = 16.67 N for each leg

T_max = η*F_N*L*sin(a) (max torque experienced in leg)

Where η is the dynamic load coefficient, F_N is normal force, L is length of lower leg segment, and a is angle it makes with normal force

T_max = 1.2*16.67*0.1397*sin(45) = **1.98 Nm**

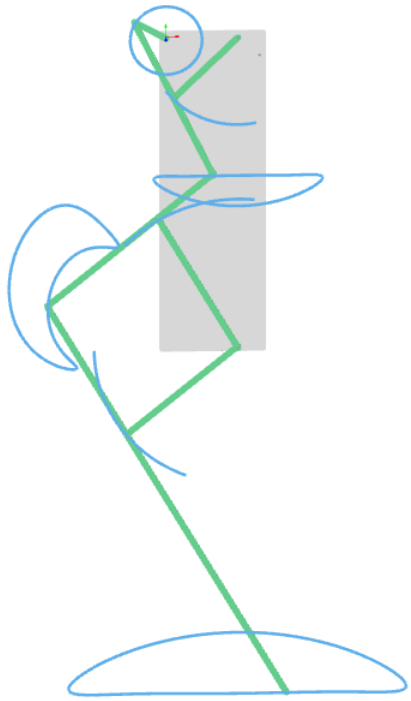


Figure C.1: Leg Motion Simulation

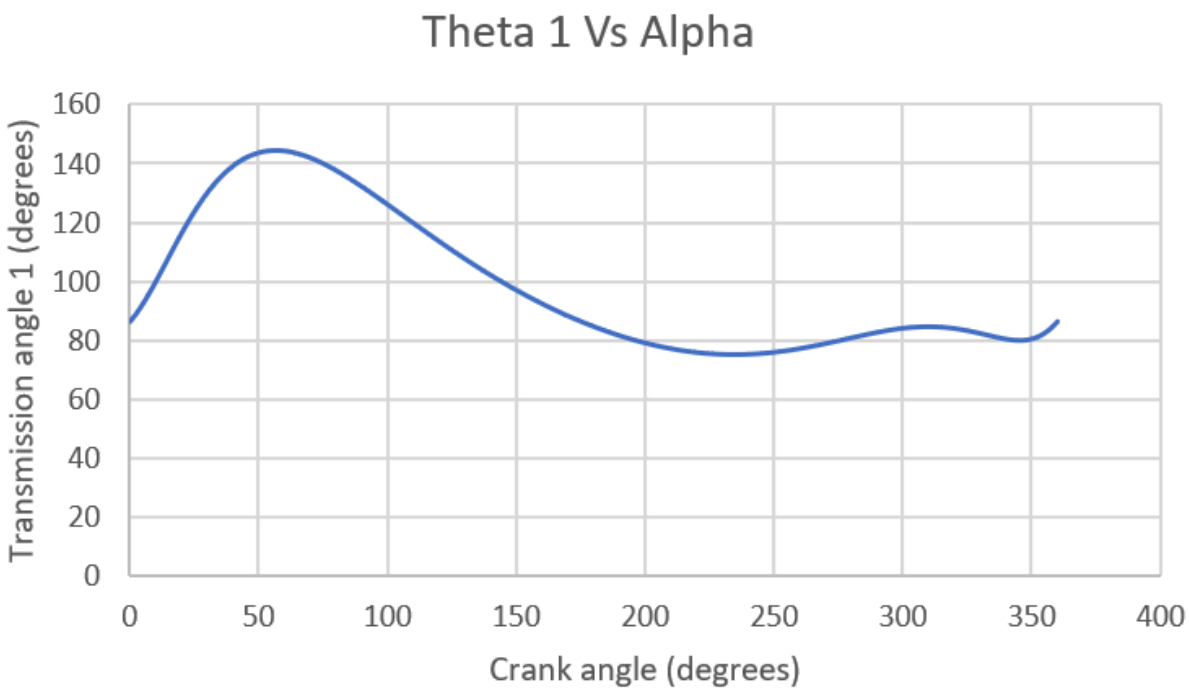


Figure C.2: 1st Transmission Angle Vs Crank Angle

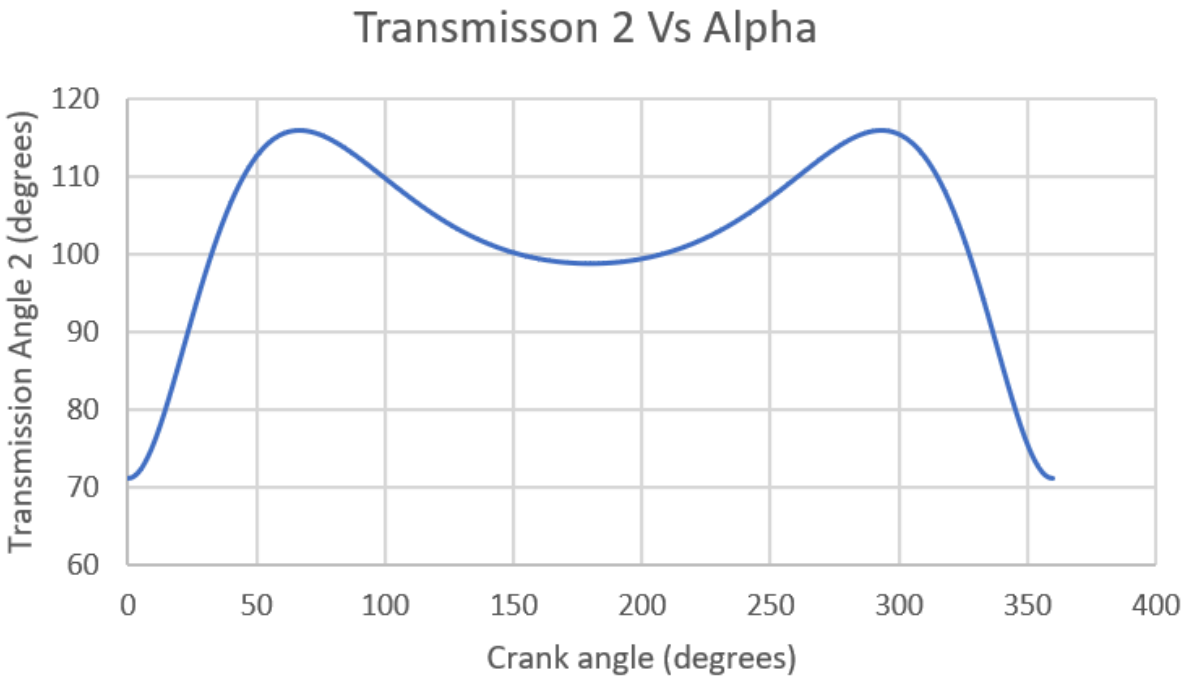


Figure C.3: 2nd Transmission Angle Vs Crank Angle

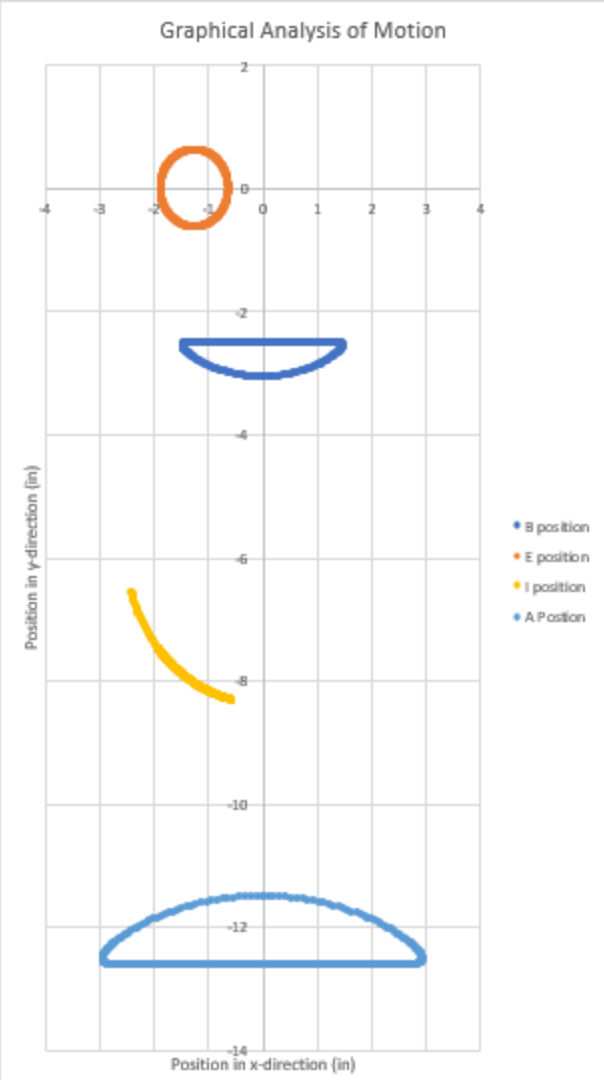


Figure C.4: Graphical Analysis of Motion Path

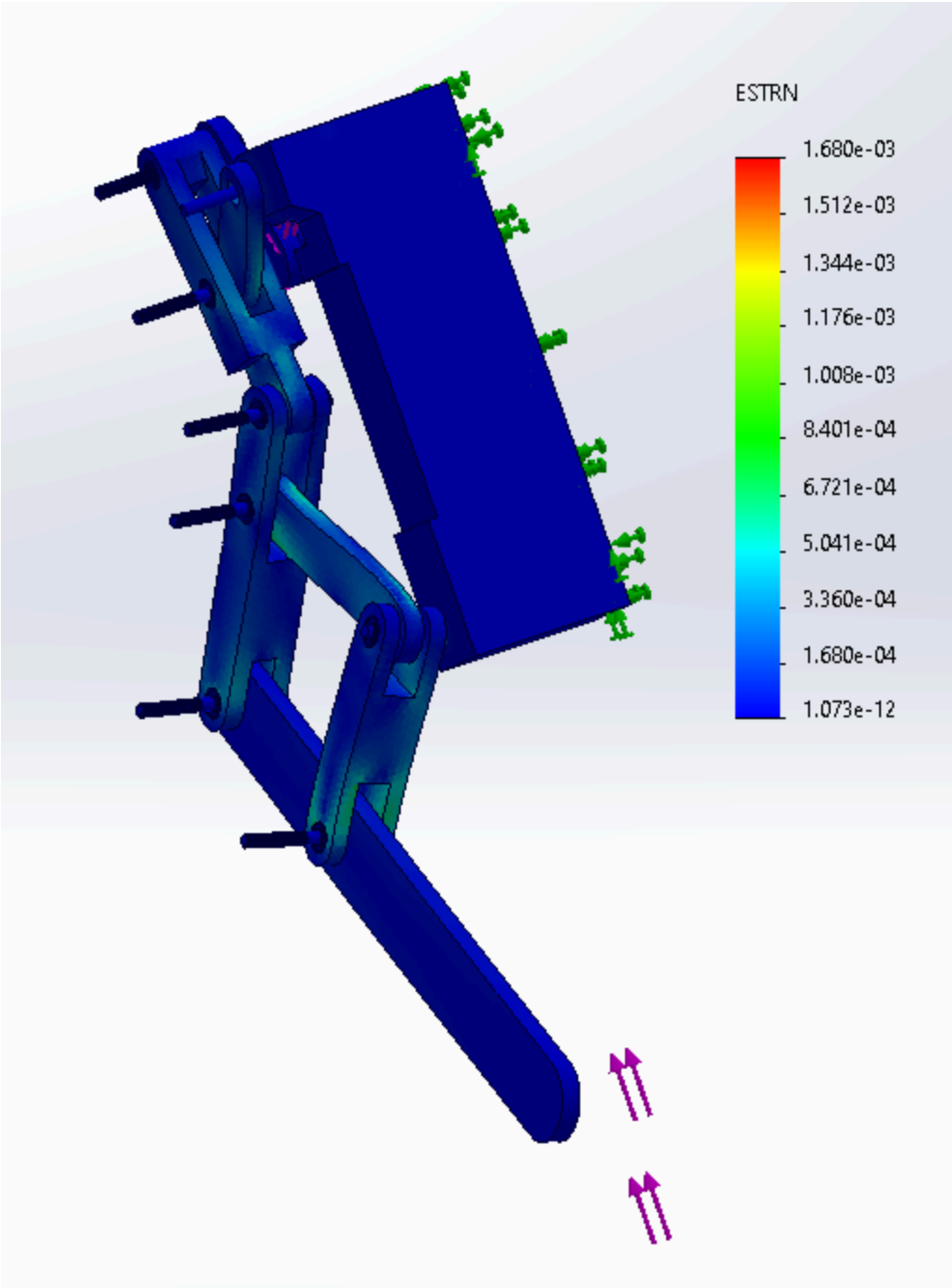


Figure C.5. Old Design strain at joints and links, exaggerated deformation

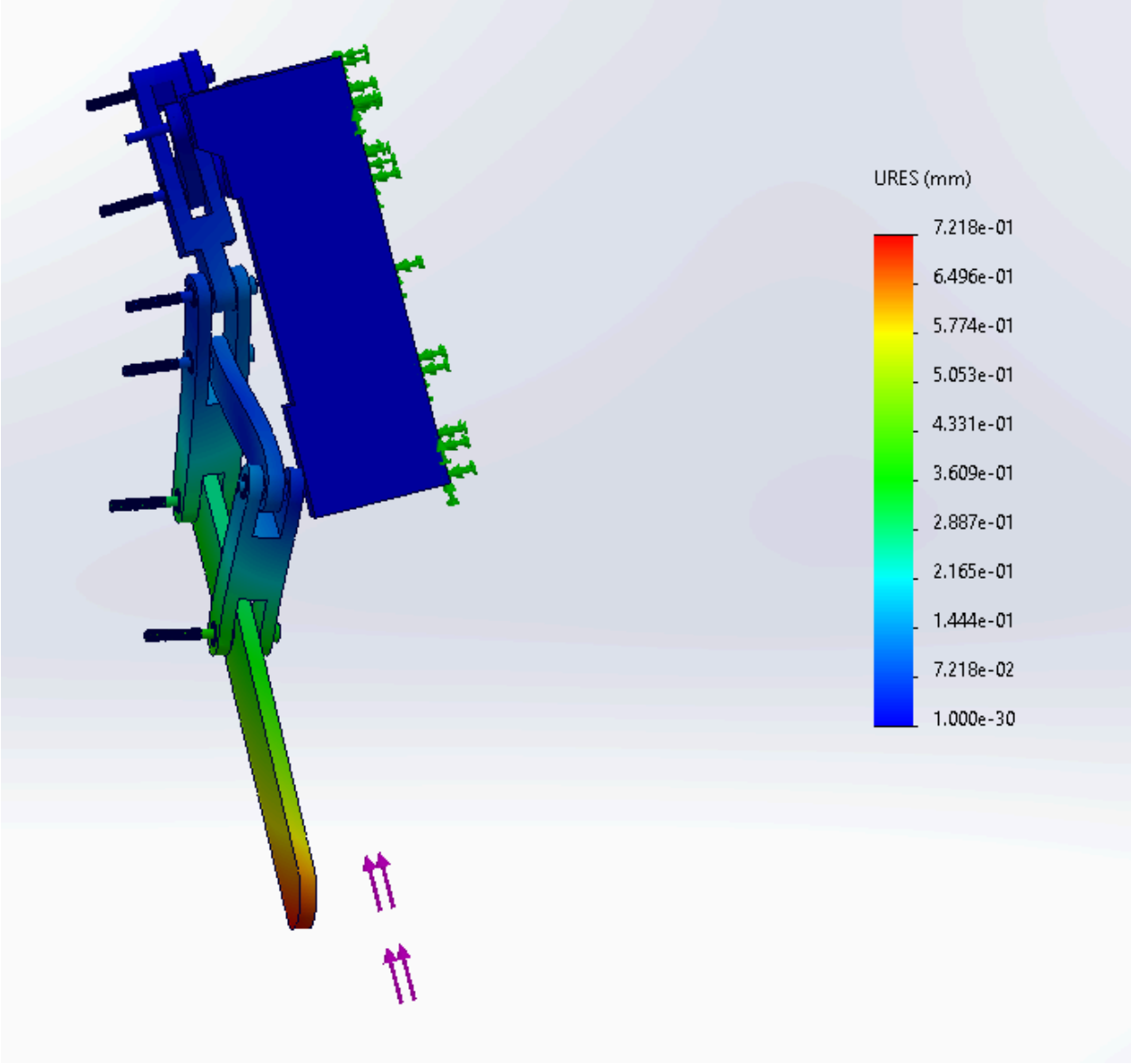


Figure C.6. Old Design Deformation study, exaggerated deformation

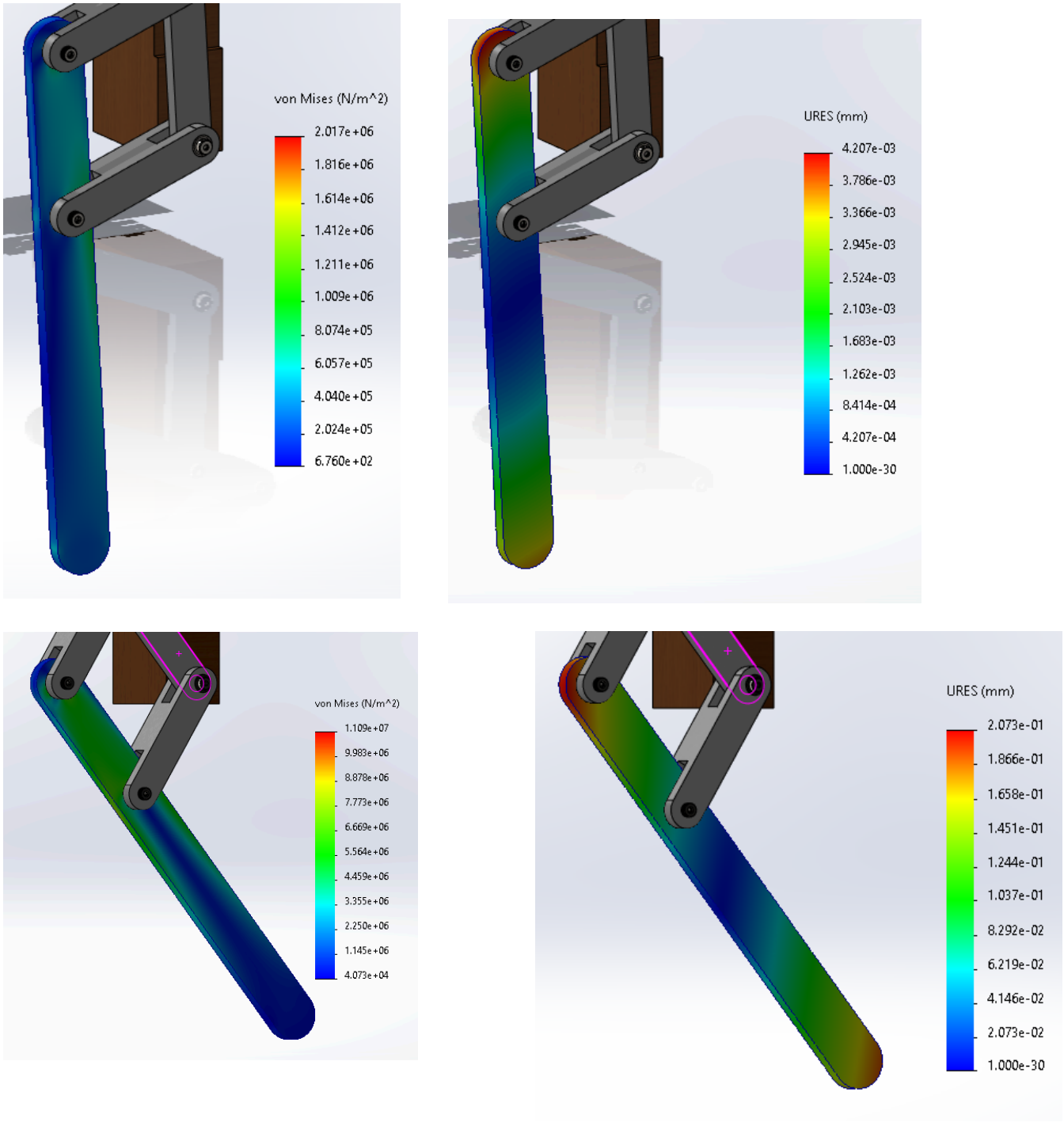


Figure C.7. Link HIA Deformation and stress from dynamic stimulation, 10 lbf max expected forces

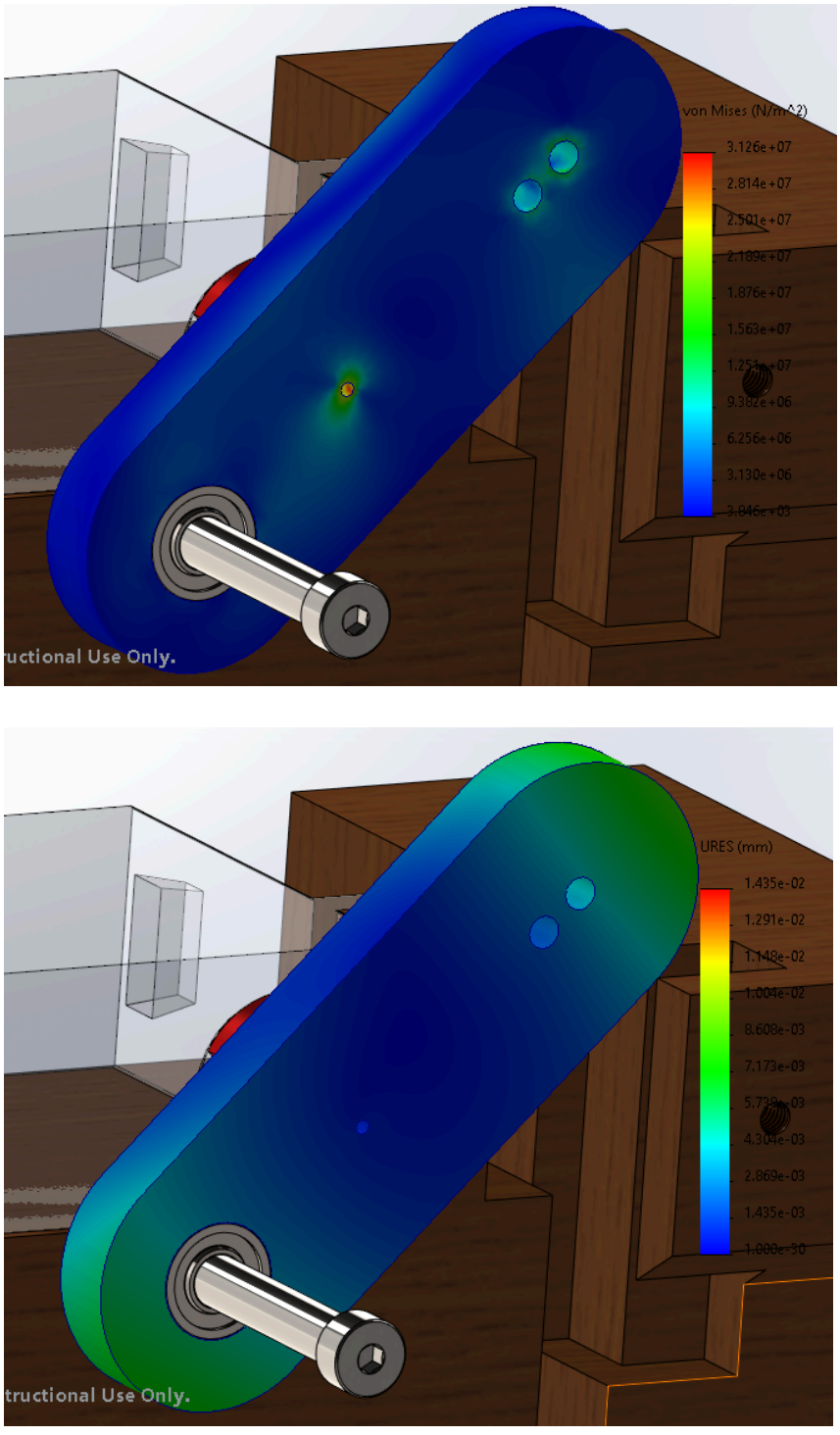


Figure C.8. Crank Stress and deformation in dynamic study

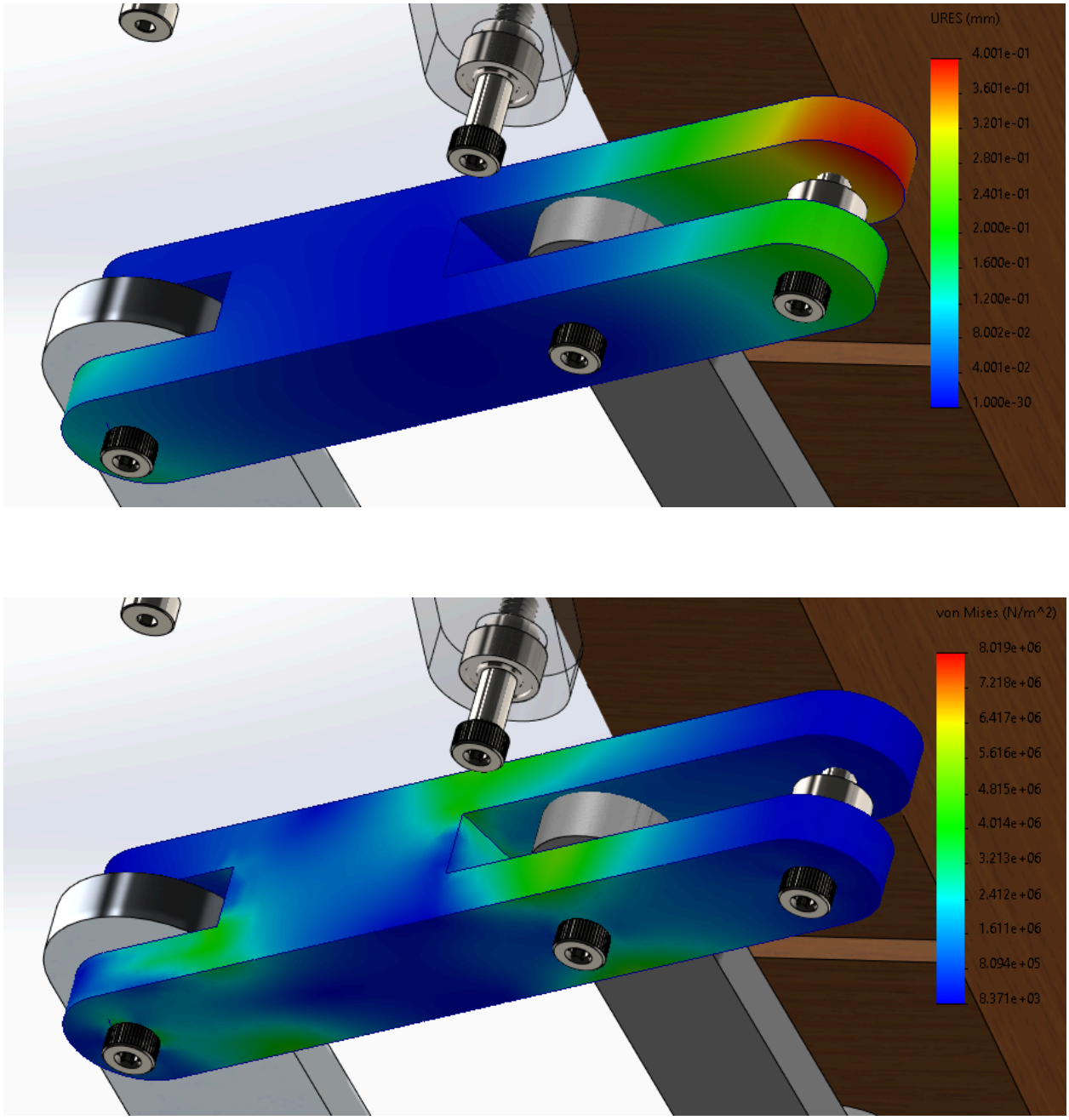
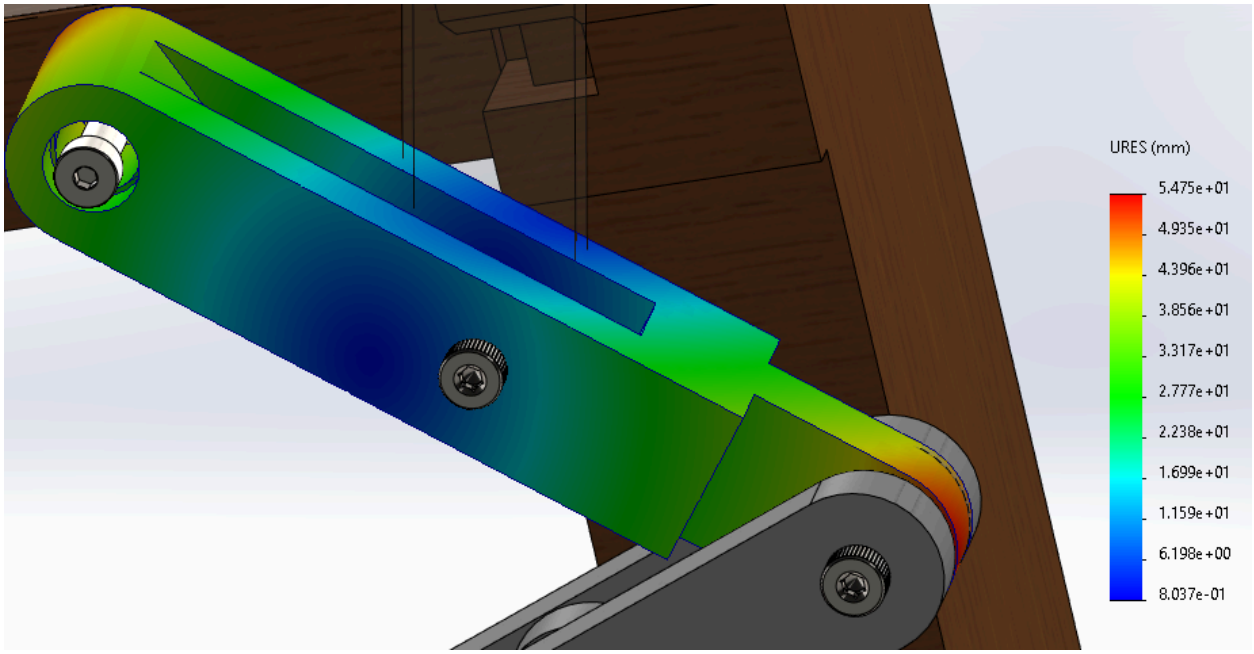


Figure C.9. Link BGH stress and deformation dynamic simulation



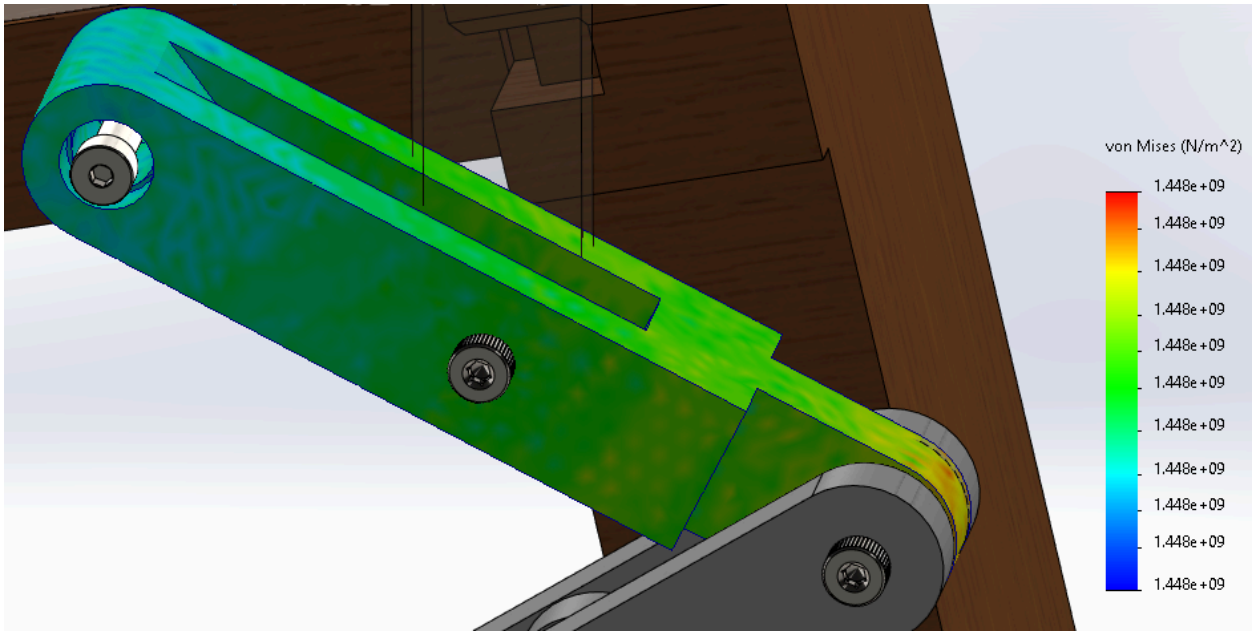


Figure C.10. Stress and deformation of link EDB if abs plastic

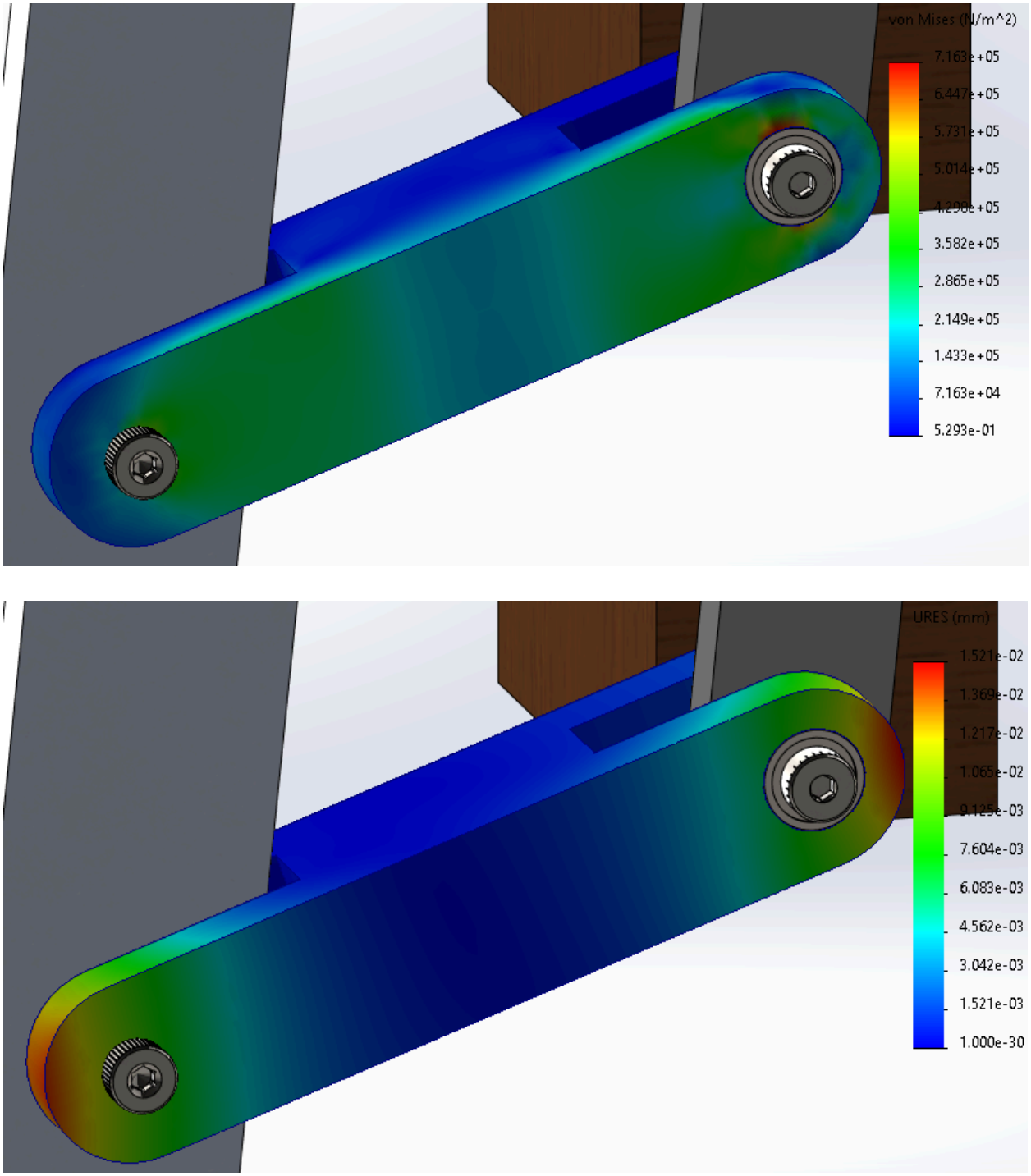
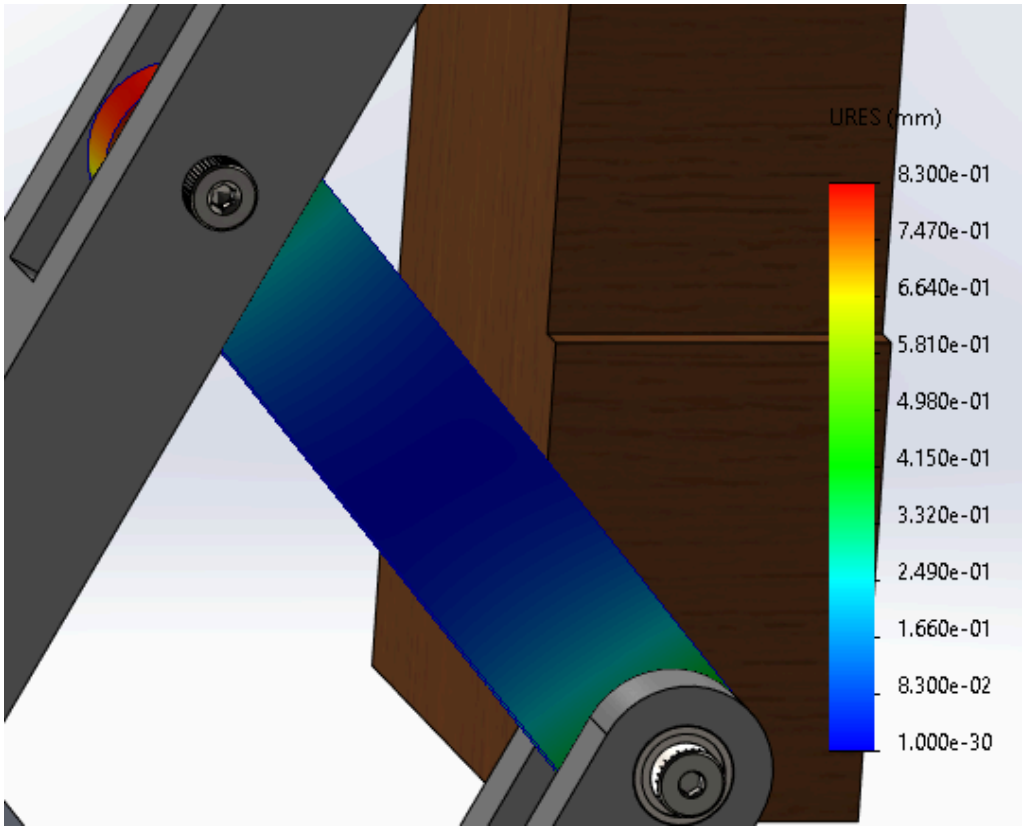


Figure C.11. Stress and deformation of link MI



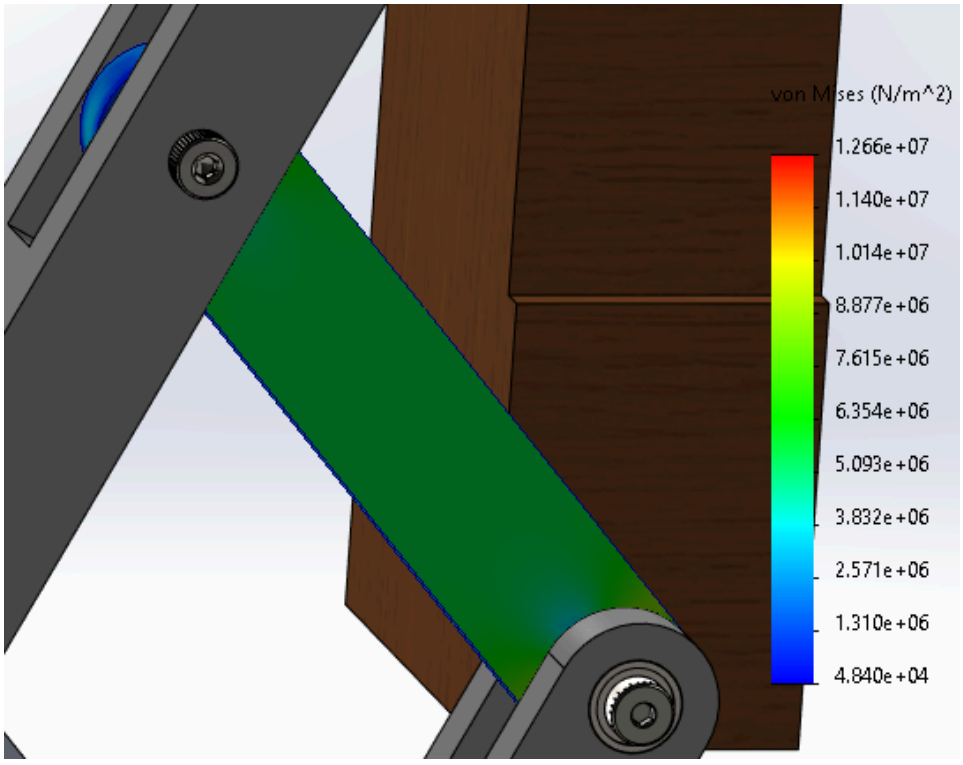


Figure C.12. Stress and deformation for link GM